Incentivizing Retail Traders: Evidence from Daily High-Water Marks on a Social Trading Platform

Justin Mohr [1] Anthony Waikel [2]

University of Alabama

October 2026

[1] University of Mississippi
[2] UMass Dartmouth

First a little background

- Anthony and I have known each other for years
  - Visiting Illinois alum Filipe Correia at UGA
- This summer I joined this paper
  - Expanding dataset to tick data
  - Doing a bit more microstructure
- Still finalizing the data collection; additional results at the end

Technology is Changing How Retail Investors Trade

- Fintech has changed how consumers interact with financial markets
  - Spending (D'Acunto et al. 2019b) and saving (Gargano and Rossi 2024)
  - Robo-advisors help investors diversify (D'Acunto et al. 2019a)
- Platform design shapes the decisions retail investors make
  - 'Gamified' interfaces alter investor attention (Barber et al. 2022; Tan 2021)
- *A growing and relatively unexplored innovation: social trading platforms*

Social Trading: Retail Traders as Fund Managers

- Retail **traders** make their investment decisions public
  - Retail **investors** buy an exchange-traded product that tracks the trader's portfolio
  - Retail traders effectively become fund managers
- The platform pays traders a performance bonus
  - A fraction of the fund's return above its previous high close
  - Earned any day the portfolio closes above that high, and the threshold resets immediately
- *An institutional high water mark contract, but paid daily to retail traders* (Ma et al. 2019; Aragon and Nanda 2012)

How Could Daily Incentives Affect Retail Traders?

- Alignment
  - Performance pay aligns trader and investor interests, potentially mitigating retail biases
    - (Jensen and Meckling 1976; Holmstrom 1979)
- Or amplification
  - Extreme frequency amplifies loss aversion, excessive trading, and short-term thinking
    - (Odean 1998, 1999; Dorn et al. 2015; Delfabbro et al. 2021)
- **This paper: How does a daily high water mark bonus change the behavior and performance of retail traders?**

Data and Methodology

- Data
  - **Social trading platform:** every trade, daily fund prices, and bonus days from 2018 to 2022
  - **Sample:** 5,147 funds and 3.6 million fund-day observations
  - **CRSP and LSEG:** returns of the underlying assets
  - *Increasing our coverage with German exchange data*
- Methodology
  - Regression discontinuity around the bonus threshold
  - Differences-in-differences around each bonus (de Chaisemartin and D'Haultfoeuille 2026)
  - January 1st reset of the high water mark
  - Counterfactual static portfolio where traders do not trade at all

Preview of the Main Results

- On days a bonus is achieved
  - Returns spike 1.6 percentage points above the previous day
    - Immediately fall back to pre-bonus levels
  - Odds of trading are 15% higher just above the bonus threshold
  - Traders sell recent winners to lock in the gain, but miss further run up
- Doing nothing would be better: 63% of funds would earn higher alpha and 57% of fund-years would earn more bonuses
- Consistent with daily incentives amplifying short-term focus rather than aligning interests

Contribution

- Performance incentives are well studied for institutional managers (Brown et al. 1996; Massa and Patgiri 2009; Agarwal et al. 2009; Ray 2009); pay can align interests (Jensen and Meckling 1976; Holmstrom 1979)
  - Daily HWM amplifies short-term focus among retail traders
- Fintech platform design can help (D'Acunto et al. 2019a; Rossi and Utkus 2021; Gargano and Rossi 2024) or hurt users (Barber et al. 2022; Tan 2021)
  - Wall Street style incentives backfire at a daily frequency
- Retail and social trading (Barber and Odean 2000; Petajisto 2013; Anginer et al. 2024; Oehler et al. 2016; Roder and Walter 2019; Oehler and Schneider 2023)
  - Retail–professional gap persists even with performance pay

Social Trading (Wikifolios)

- Individuals register and build a virtual portfolio using over ∼600,000 financial products, including
  - Stocks
  - Exchange-traded products
  - Funds (real estate, bonds, equities, money market, or mixed funds)
  - Investment certificates (index certificates, bonus certificates, discount certificates, and reverse convertibles)
  - Leverage products
- Traders trade virtually, but prices reflect real-time bid/ask quotes
- An index is created to replicate the virtual portfolio's performance 1:1

Starting a Wikifolio

- Any registered user can create a wikifolio: a virtual portfolio plus a written trading idea (at least 200 characters)
- The trader chooses the investment universe the wikifolio can trade
- **Test** status: private, only the trader can see it
- To publish: trader profile, title, idea, at least one trade, and verified email
- **Published** status: searchable, users can add it to their watchlist and follow every trade

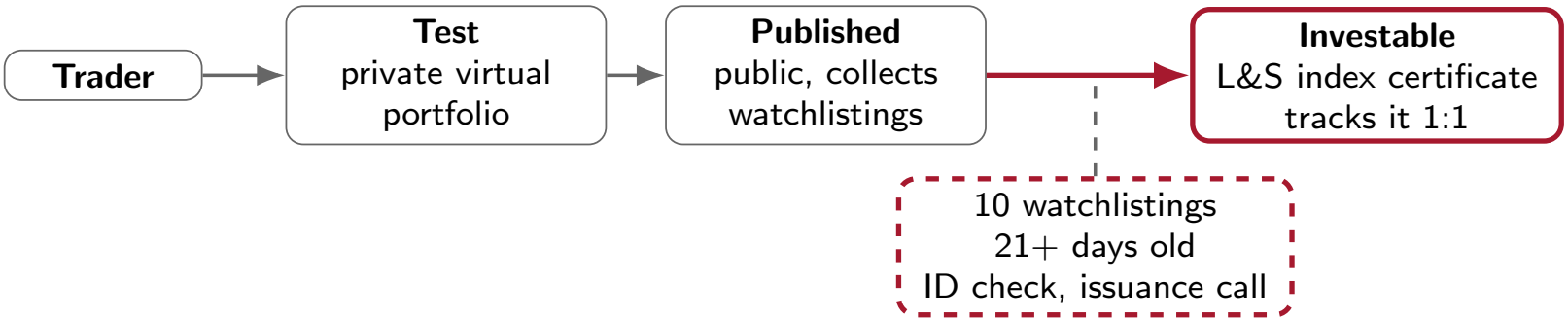
How a Wikifolio Works



Becoming Investable

- At least 10 watchlistings and the wikifolio is at least 21 days old
- Legitimation: ID, address, and phone number (18 or older)
- Issuance call with wikifolio about the trading idea and strategy
- Lang & Schwarz issues an open-ended index certificate with its own ISIN that tracks the wikifolio 1:1
- Once issued, the investment universe is fixed

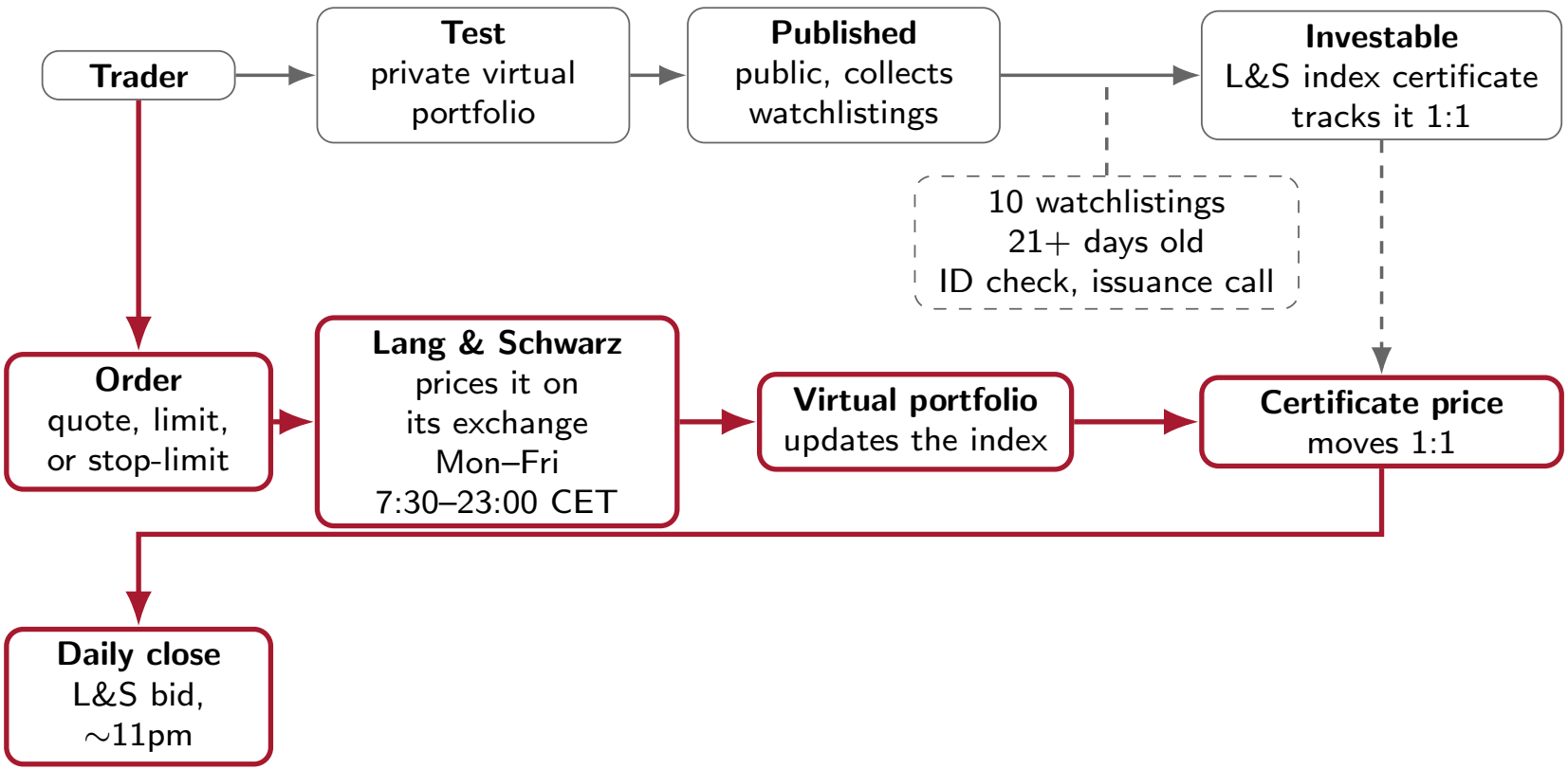
How a Wikifolio Works



Trading

- Trades are virtual, priced at real-time quotes for the underlying asset on the exchange run by Lang & Schwarz
  - Quote orders (price valid for 3 seconds), limit orders, and stop-limit orders
- Trading hours (CET): Mon–Fri 7:30am–11pm, Sat 10am–1pm, Sun 5–7pm
- No transaction costs for trades inside the wikifolio
- With invested capital, Lang & Schwarz hedges the positions
- Daily close = Lang & Schwarz bid price at about 11pm

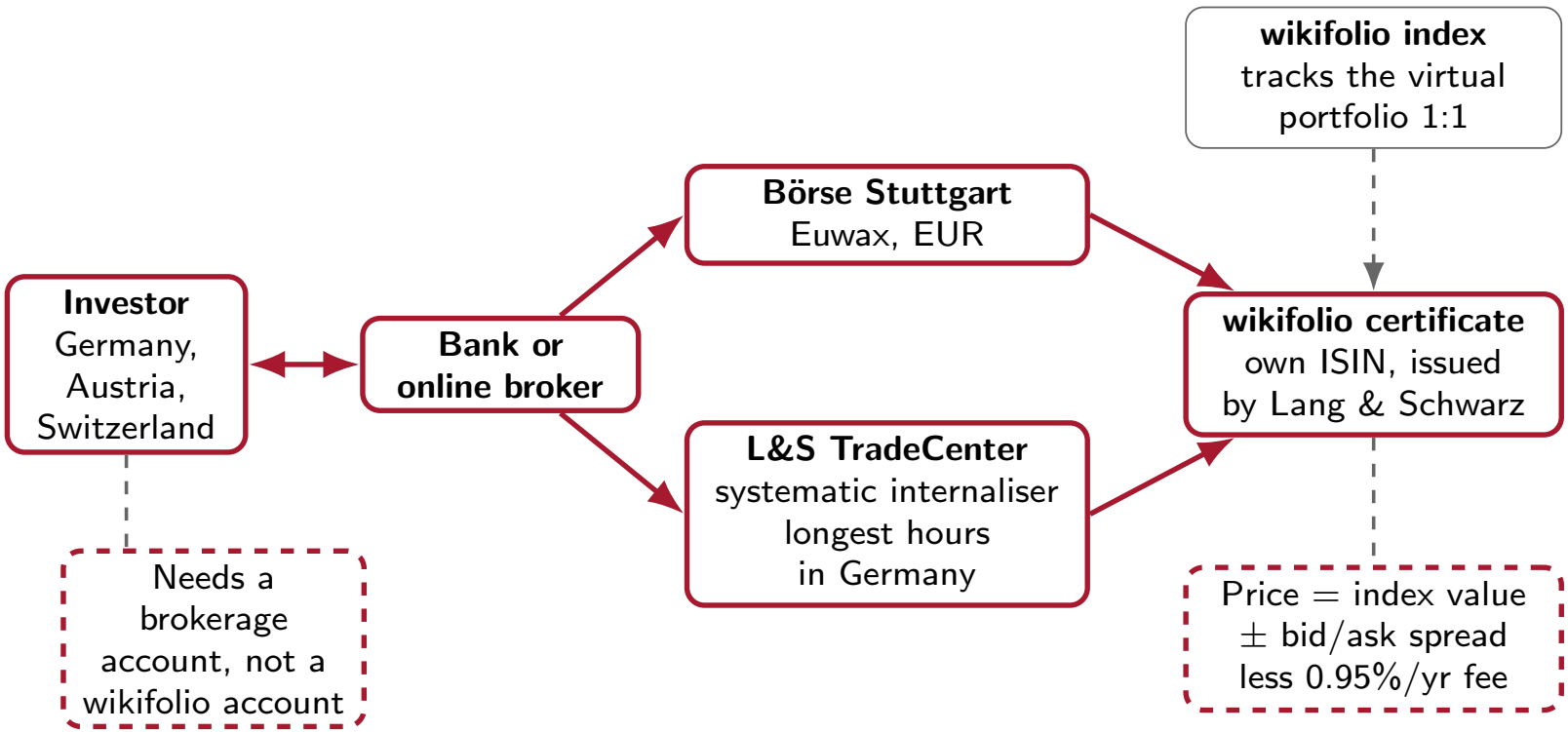
How a Wikifolio Works



Investing in Wikifolio Certificates

- Investors buy the certificate by its ISIN through their own bank or online broker, with no wikifolio account needed
- The certificate is not traded on the Lang & Schwarz exchange, but on
  - Börse Stuttgart (Euwax, in EUR), or
  - Directly with Lang & Schwarz TradeCenter: a systematic internaliser, Germany's off-exchange venue with the longest trading hours
- Who can buy
  - Germany: every certificate is publicly offered
  - Austria: certificates of Austrian traders or with high demand
  - Switzerland: only stock-only wikifolios (BX Swiss, in CHF)
- Price tracks the wikifolio 1:1, with a bid/ask spread from its holdings and a 0.95% annual certificate fee charged daily

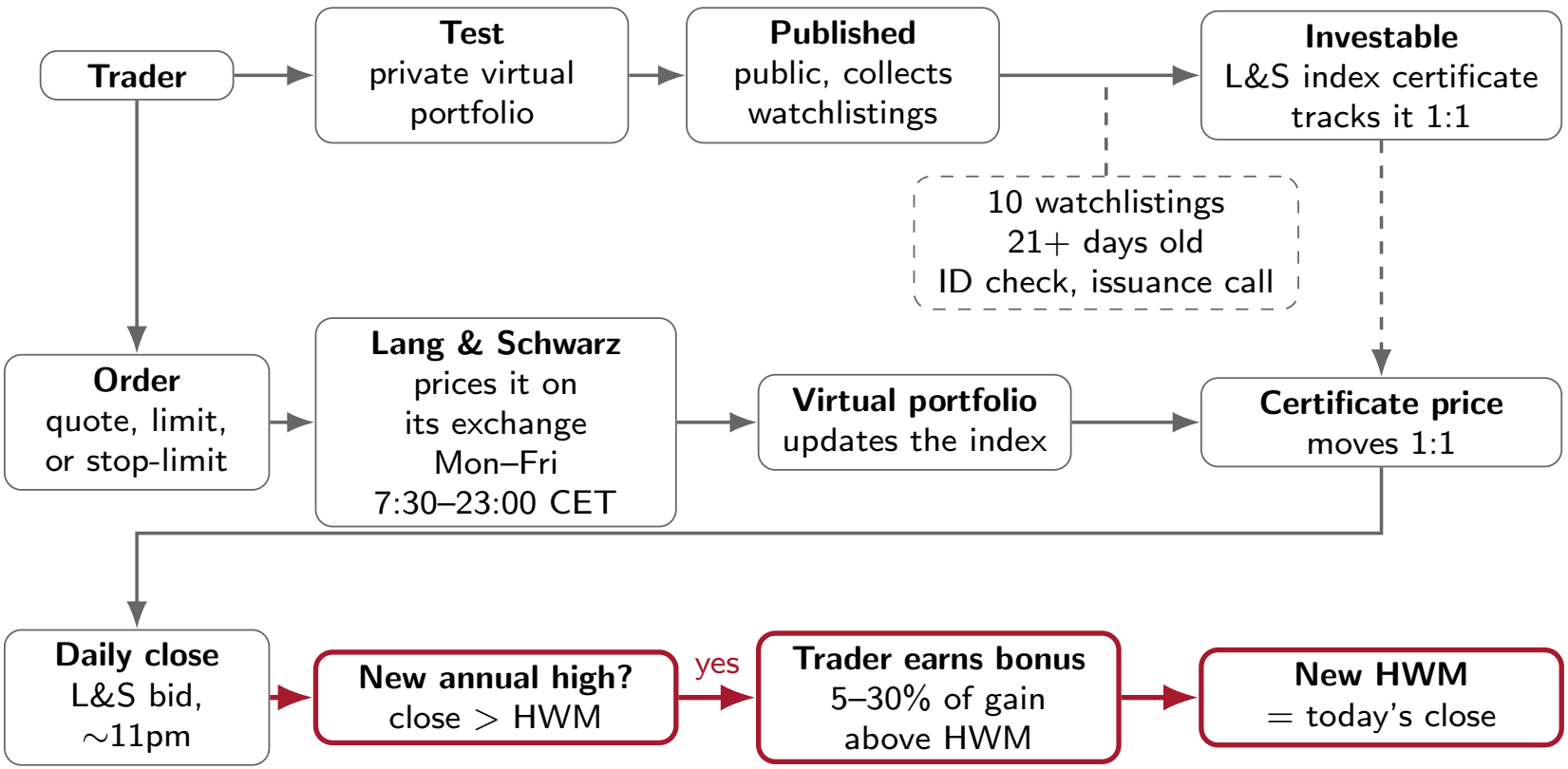
How Investors Buy In



Fees

- Traders earn no fee on assets under management, only performance bonuses
- Bonus rate of 5–30% of the gain above the high water mark (HWM), set by the trader at launch and fixed
- Earned only when the daily close sets a new annual high
- The HWM resets every January 1st

How a Wikifolio Works



Bonus Calculation

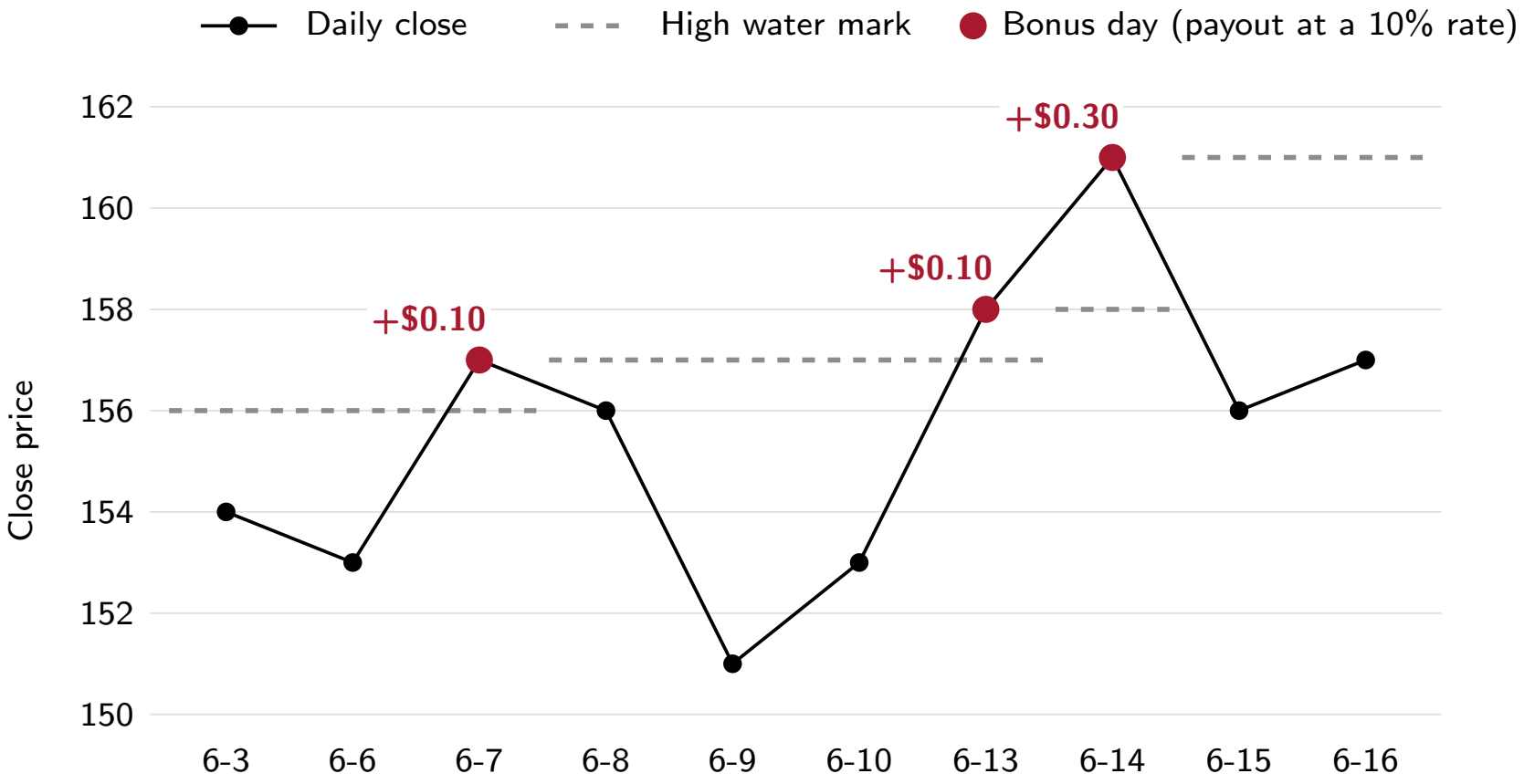
$$Bonus_{i,t} = \begin{cases} (Price_{i,t} - HWM_{i,t}) * Rate_i, & \text{if } Price_{i,t} > HWM_{i,t} \\ 0, & \text{otherwise} \end{cases}$$

Sample High Water Mark

Date	Close Price	High Water Mark	Bonus Earned?
6/3/22	154	156	No
6/6/22	153	156	No
6/7/22	157	156	Yes
6/8/22	156	157	No
6/9/22	151	157	No
6/10/22	153	157	No
6/13/22	158	157	Yes
6/14/22	161	158	Yes
6/15/22	156	161	No
6/16/22	157	161	No

Close Price = Lang & Schwarz bid at ∼11pm CET

Sample High Water Mark



Bonus = (Close − HWM) × Rate

Data

	Current	**Extension**
Period	2018–2022	Through 2026
Underlying returns	Daily returns from CRSP and LSEG	Tick-level exchange and OTC tape data, quotes and trades (finalizing collection)

- **Filters:** drop closes below \$1, each fund's first month, January, and days over \$100 from a bonus; keep funds with 20+ trades a year and 6+ active months
- **Current sample:** 5,147 funds and 3.6 million fund-days (from 8,155 funds)

Portfolio Characteristics

	Mean	Median	Standard Deviation
Daily Trades	1.17	0	4.94
Daily Buys	0.66	0	3.02
Daily Sales	0.51	0	2.49
Daily Trades when Trades $>$ 0	5.43	3	9.48
Daily Quantity when Trades $>$ 0	5,206	100	349,310
Daily Dollar Amount when Trades $>$ 0	\$24,150	\$2,086	\$436,231
Portfolio Life (Years)	2.65	2.37	1.49
Daily Returns	0.02%	0%	2.05%
Daily Alpha	-0.0002%	-0.0001%	0.37%
Performance Fee	10.78%	10%	6.29%
Bonuses Earned Per Year	30.1	20	32.6

Fund size is set by the trader at creation

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Two snapshots of open wikifolios

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67% of funds exhibit a negative alpha

Portfolio Characteristics

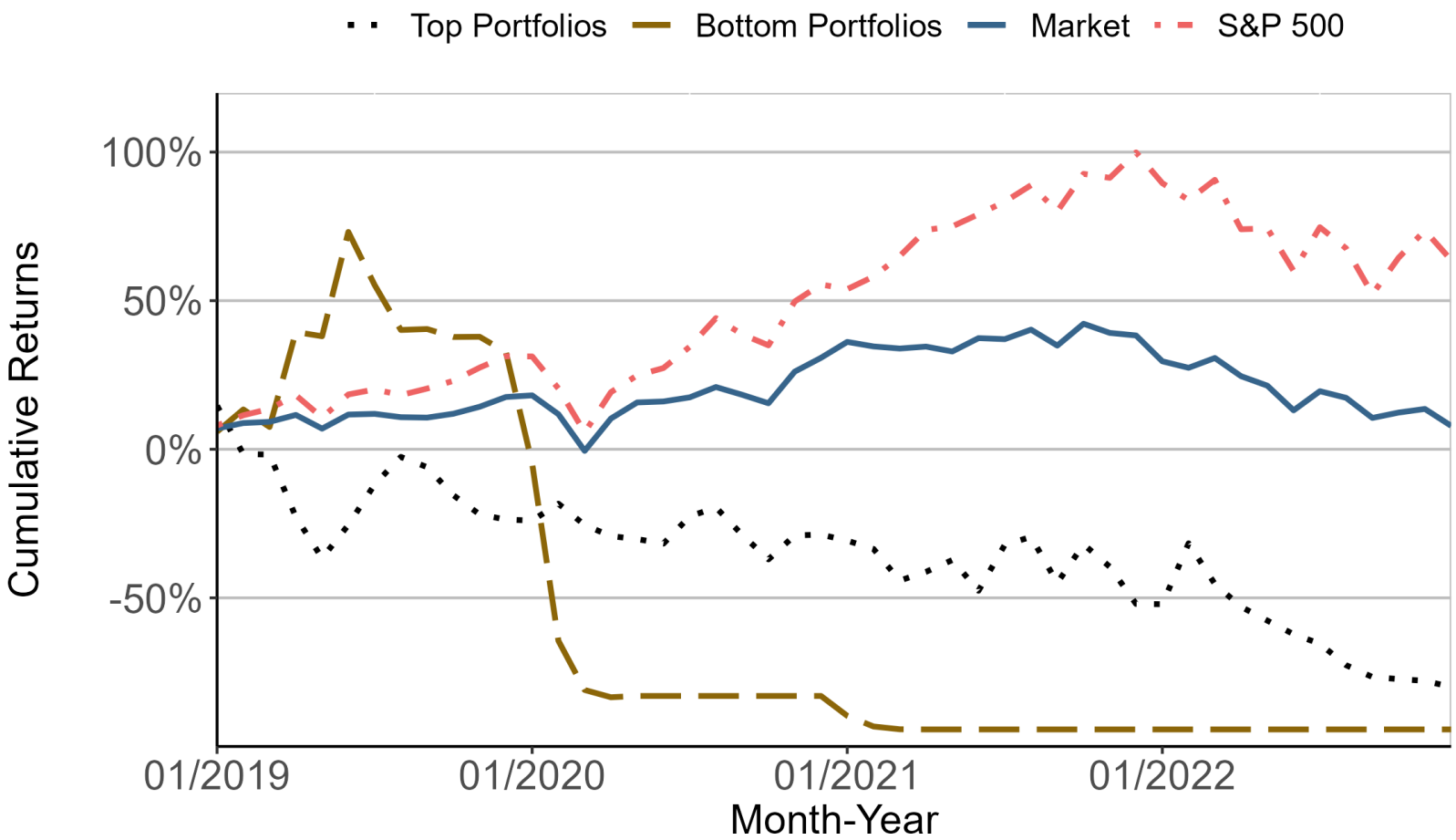
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Chosen by the trader, ranging from 5% to 30%

Return Benchmarking

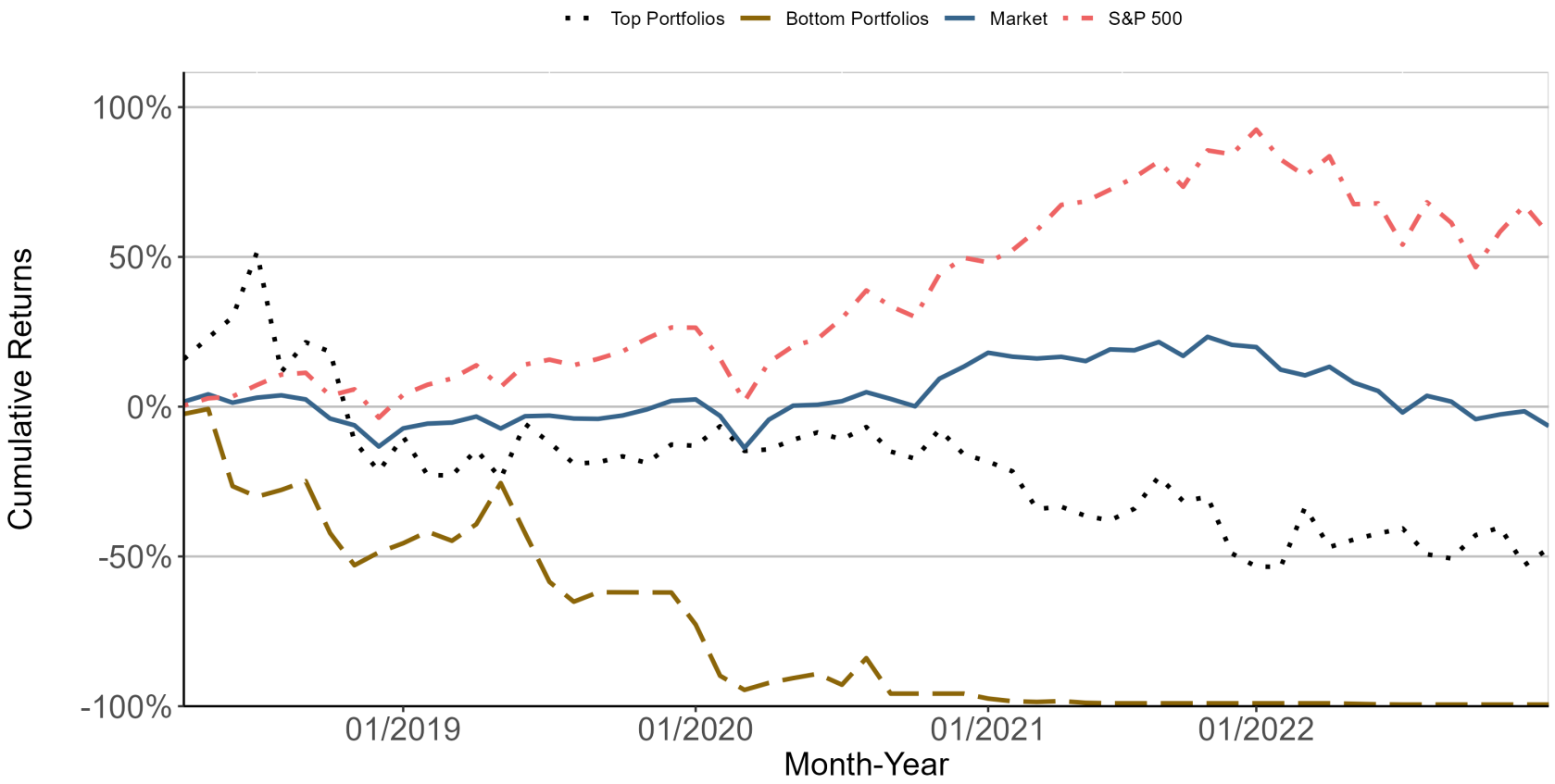
- Exercise
  - Rank funds by average return over the prior period
  - Invest in the ranked funds and hold for one period
  - Re-rank and rebalance either yearly, quarterly, or monthly
- Portfolios compared
  - **Top Portfolios:** 10 funds with the highest prior-period return
  - **Bottom Portfolios:** 10 funds with the lowest prior-period return
  - **Market:** equal-weighted, every fund active that period
  - **S&P 500:** benchmark, since 68% of funds trade US assets

Yearly Rebalancing



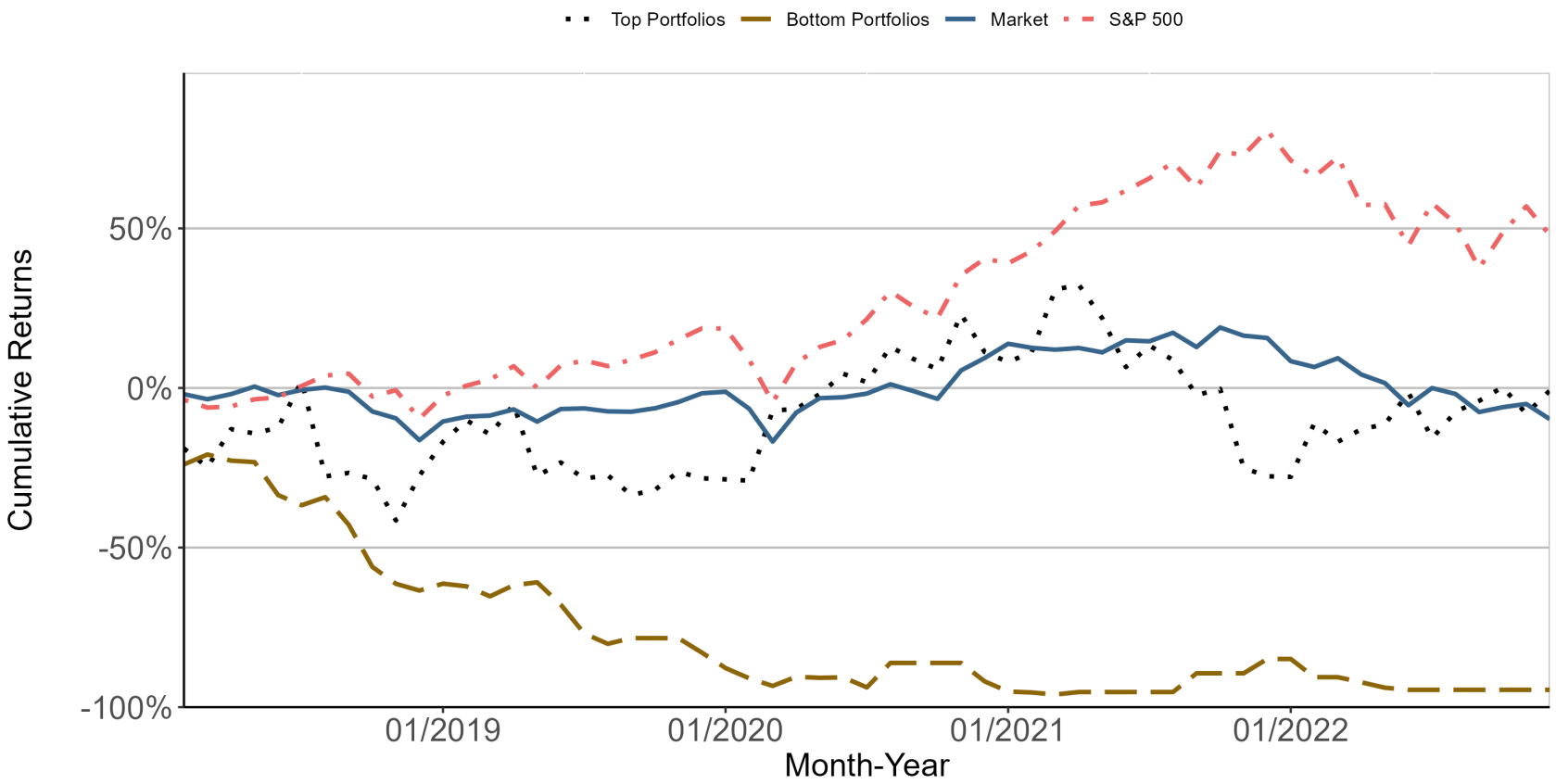
Both the top and bottom funds yield almost complete losses

Quarterly Rebalancing



Slight improvement for the top funds, which only lose about 50%

Monthly Rebalancing

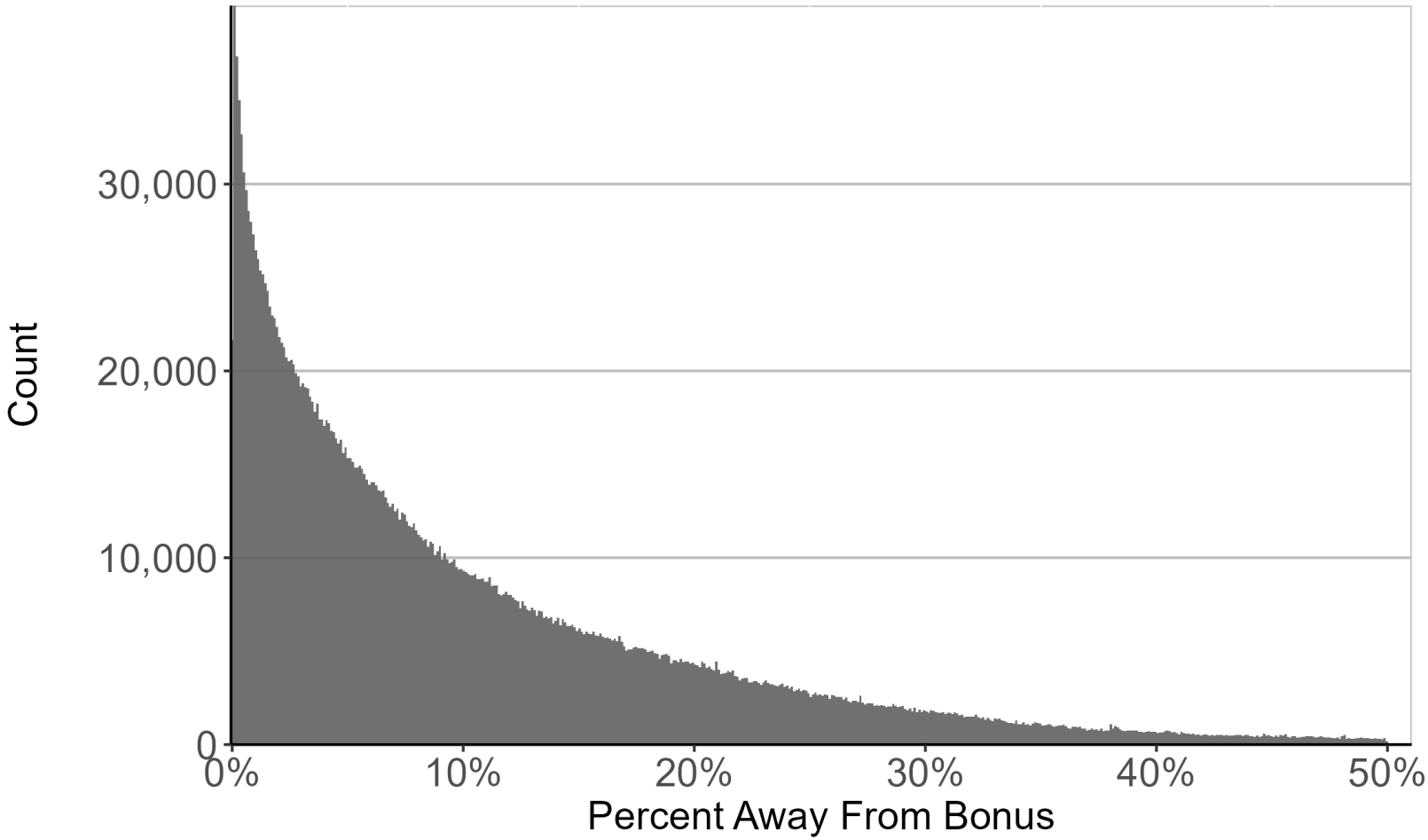


Top performers break even, but no consistent performance

Why Doesn't Performance Persist?

- In line with traders favoring short-term strategies, rather than buy and hold
  - Most traded: leveraged ETFs on market indices, and stocks like Apple, Tesla, Amazon, and Nvidia
- Daily HWM could encourage this
- **How close are traders to a bonus on a typical day?**

Distribution of Distance to Bonus

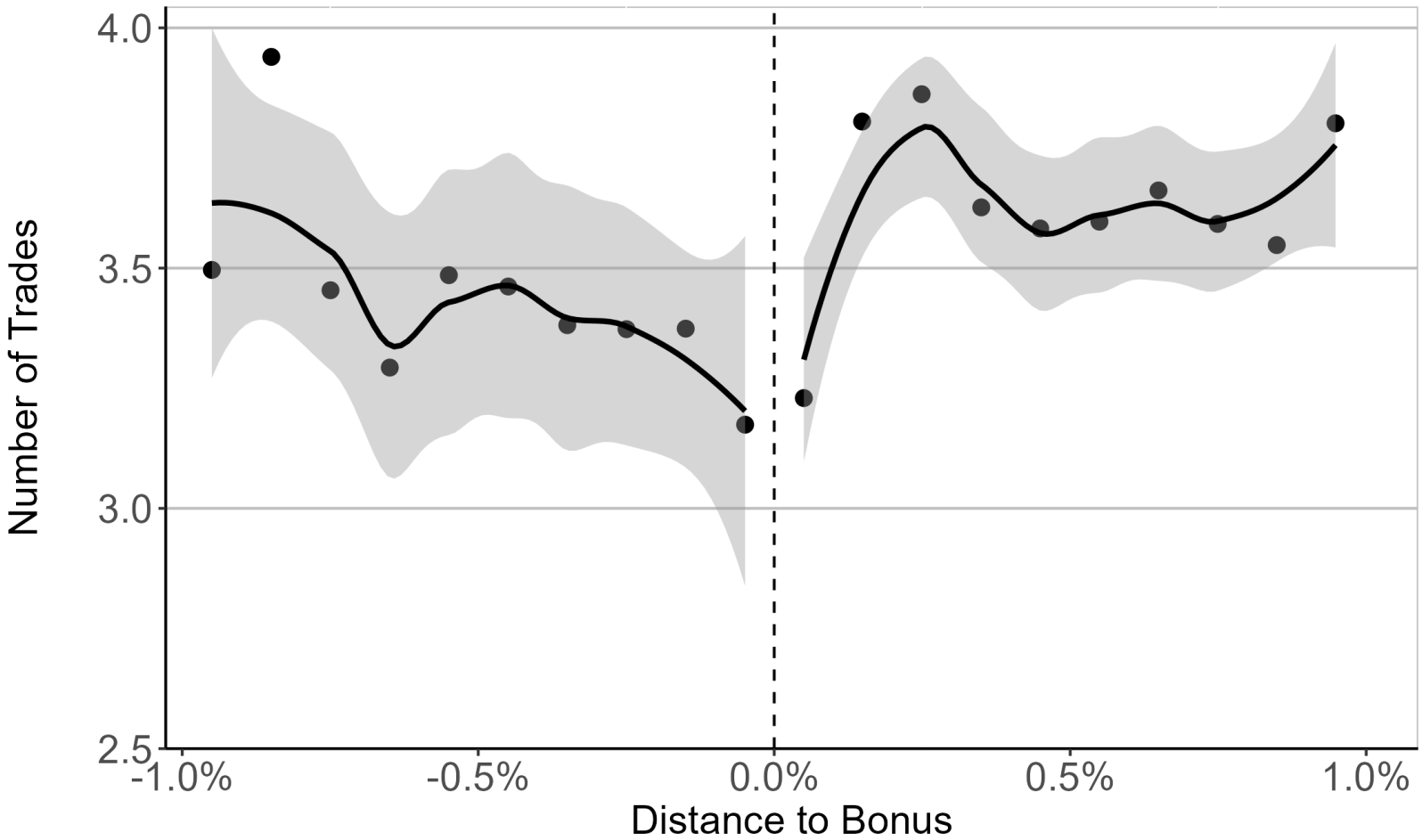


39% of all trader-days are within 5% of the bonus threshold

Is the Bonus Salient?

- Regression discontinuity around the bonus threshold
- Compare trader-days just above vs. just below the threshold (within $\pm 1\%$)
- Outcomes over the prior 3 trading days (traders average about 1 trade a day):
  - Number of trades
  - Whether any trade was made

RDD Around Bonus



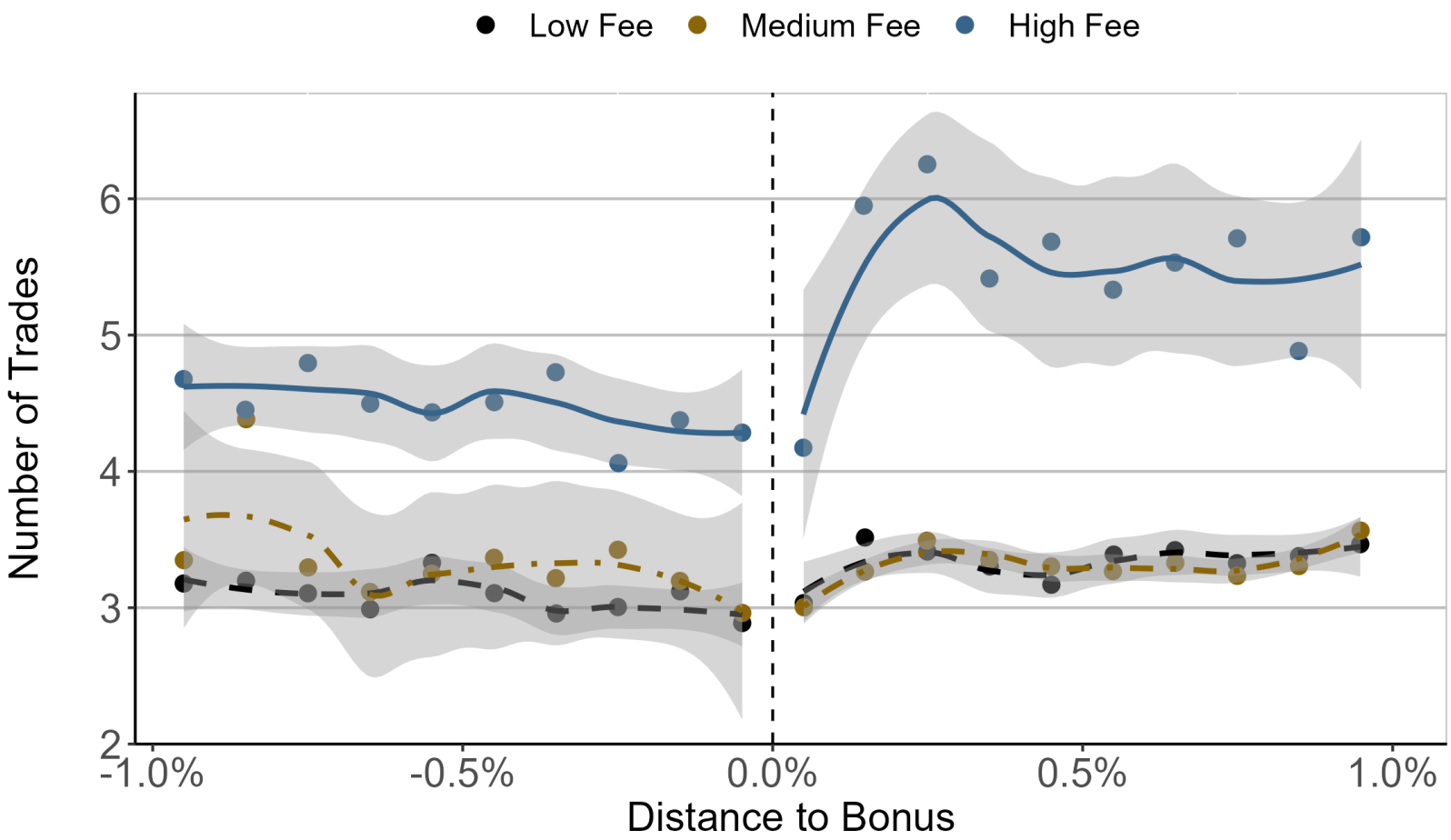
Intensive: 0.31 trades (9% of mean) | Extensive: odds increase 15%

▸ Regression Table

Bonus Fee Groups

- The trader picks a 5–30% bonus rate when creating the fund, and it cannot be changed
- Three groups:
  - Low: $\leq 7.5\%$ (1,881 traders)
  - Medium: 7.5–15% (2,375 traders)
  - High: $> 15\%$ (891 traders)

RDD Around Bonus By Group



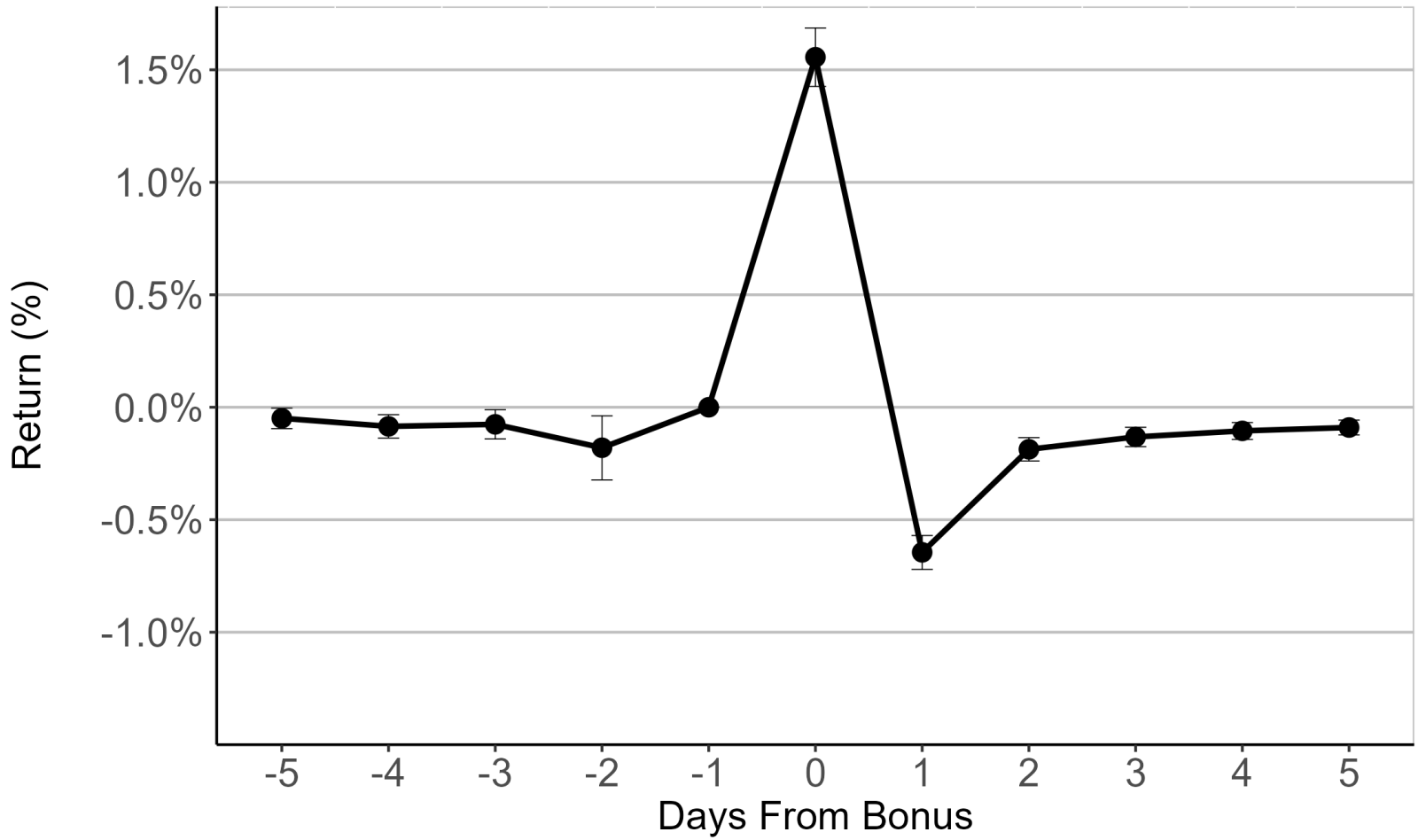
Higher bonus motivates greater trading around the HWM

Differences-in-Differences Estimator

- Treatment = earning a bonus, which is
  - **Non-absorbing:** the HWM resets every January 1st
  - **Non-binary:** traders earn many bonuses a year
- Two-way fixed effects is biased with heterogeneous, staggered effects (Callaway and Sant'Anna 2021; Sun and Abraham 2021)
- Estimator of de Chaisemartin and D'Haultfoeuille (2026): change from the day before a bonus, vs. funds with the same treatment history not yet treated

$$DID_{g,\ell} = Y_{g,F_{g-1}+\ell} - Y_{g,F_{g-1}} - \frac{1}{N_{g,F_{g-1}+\ell}} \sum_{g':D_{g',1}=D_{g,1},F_{g'}>F_{g-1}+\ell} \left[Y_{g',F_{g-1}+\ell} - Y_{g',F_{g-1}}\right]$$

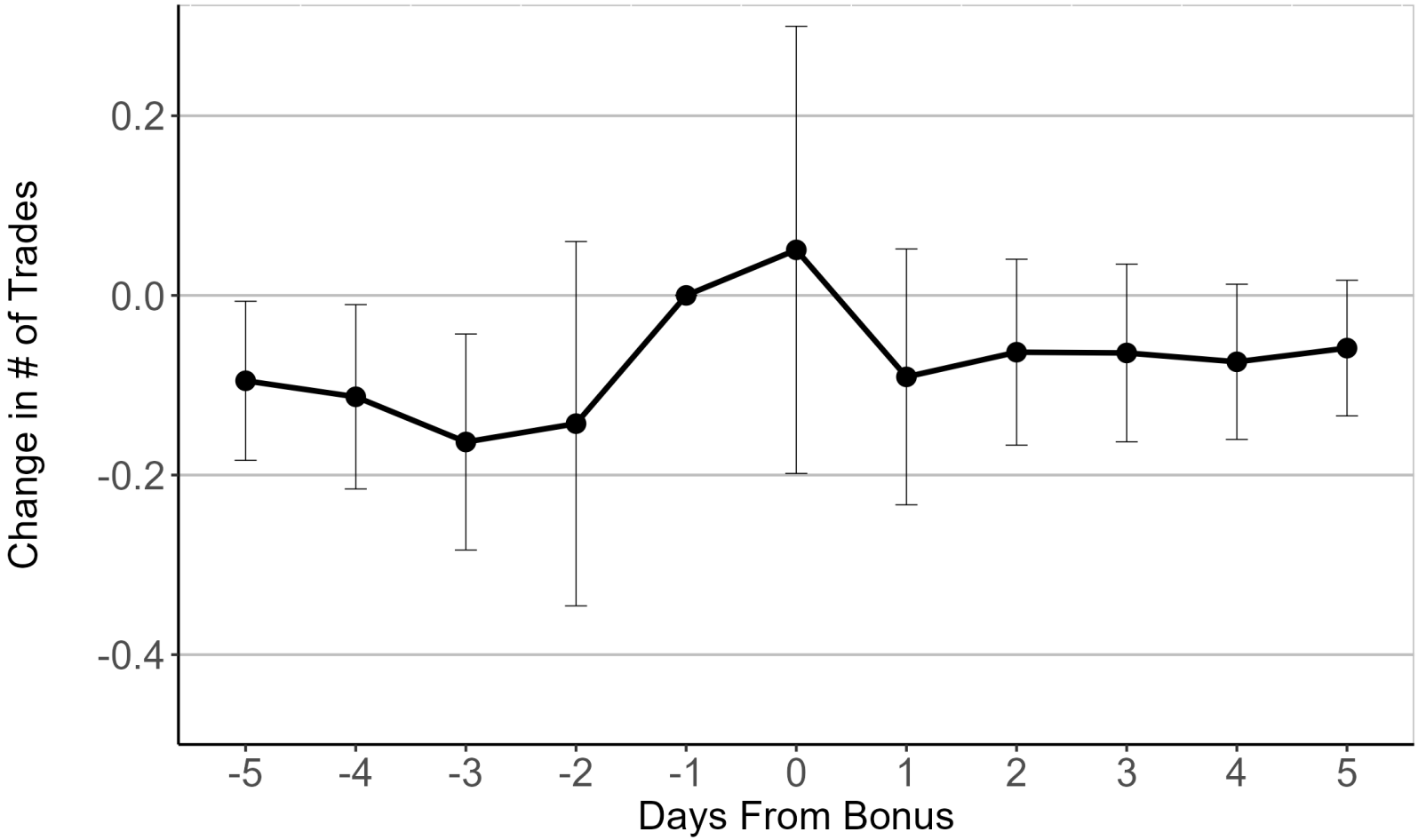
Returns Around Bonuses



Traders earn a 1.6 pp higher return on the bonus day

▸ Regression Table

Number of Trades Around Bonuses



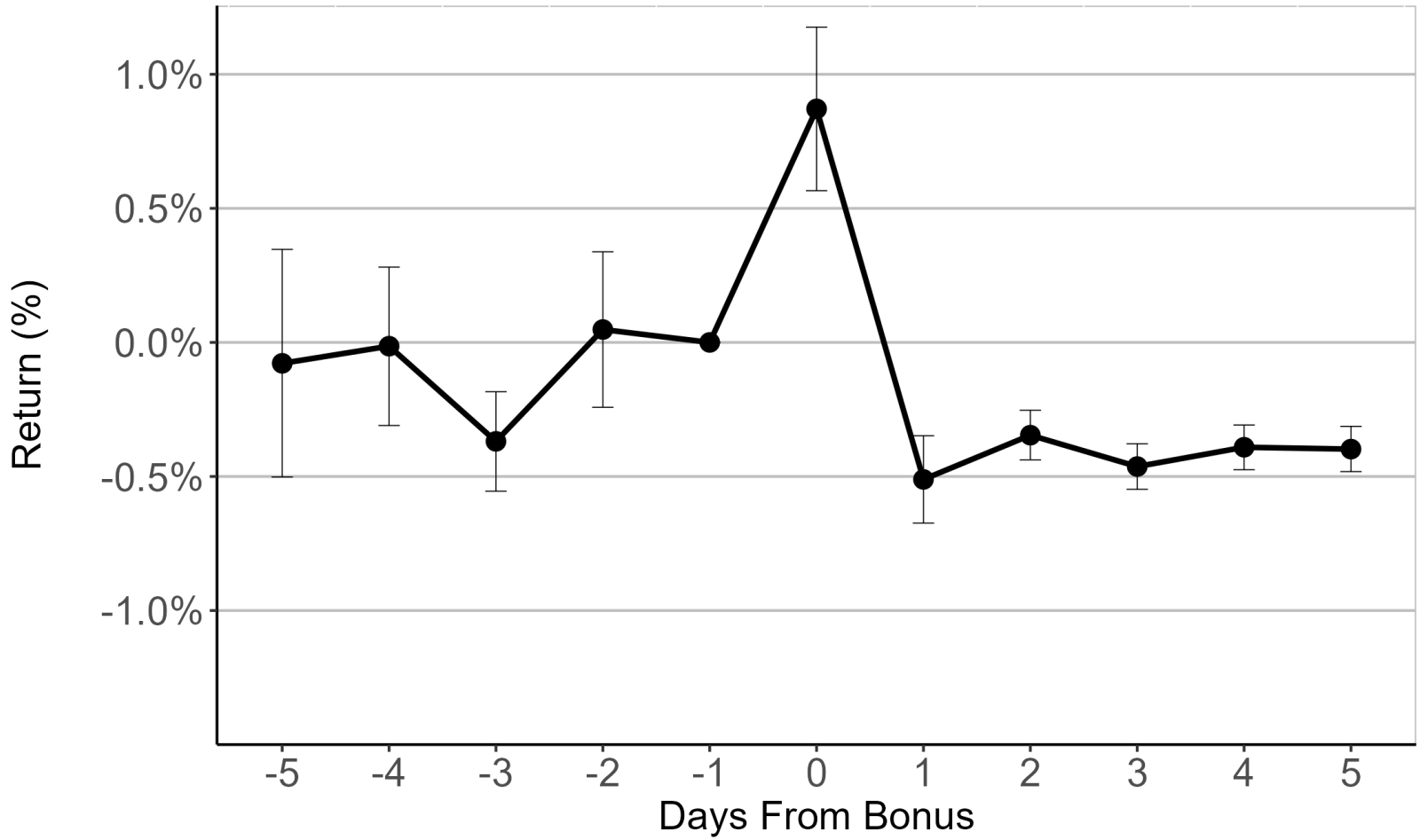
No significant increase in trades placed

▸ Regression Table

January Reset

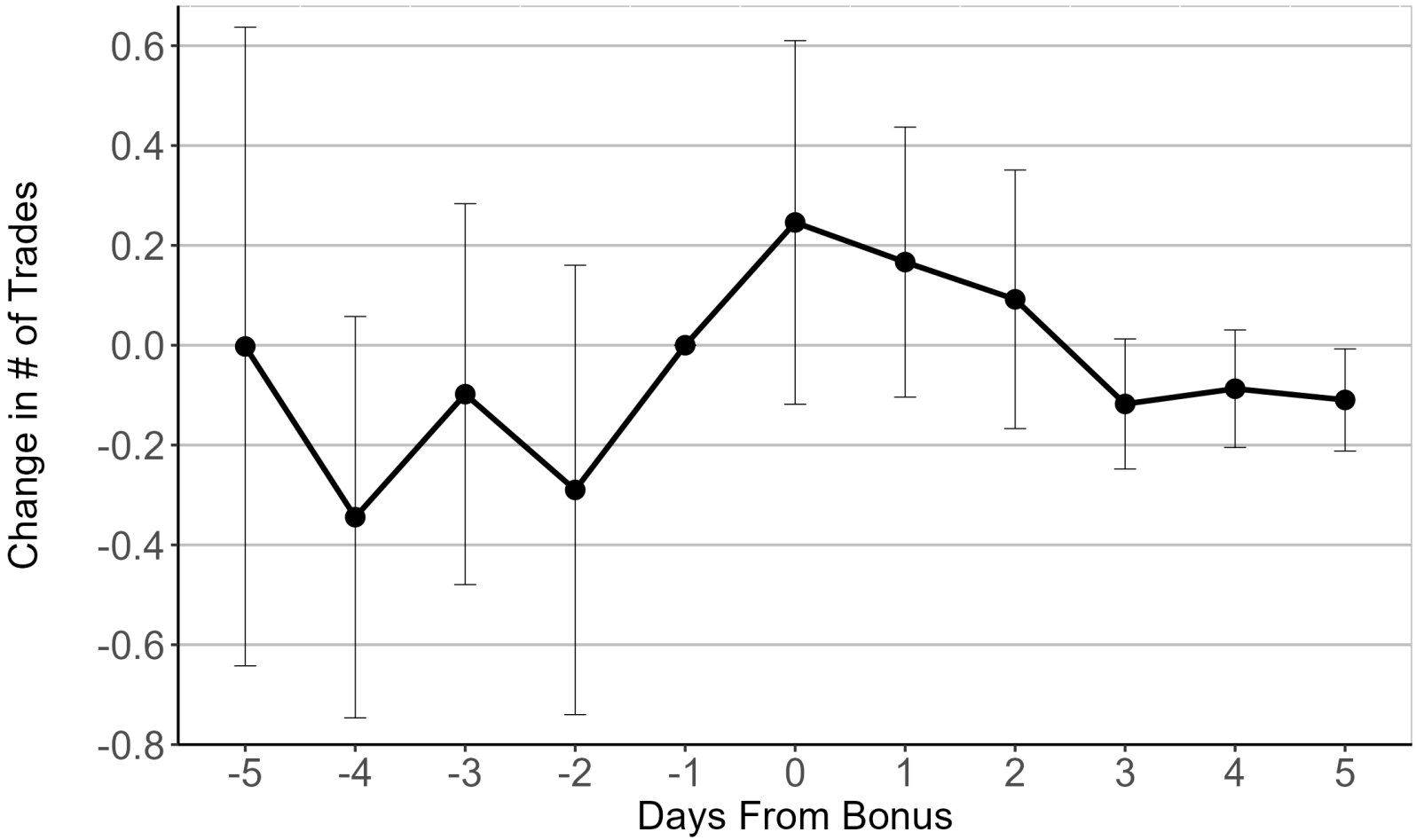
- Concern: more skilled traders may cross earlier or more often, and crossing depends on past performance
- Every January 1st all HWMs reset to that day's close
  - More easily achievable bonus allows us to isolate its effect
- Re-run the DiD on January only, first 3 bonuses per trader (437,046 observations)

January: Returns Around Bonuses



Similar pattern to the whole sample

January: Number of Trades Around Bonuses

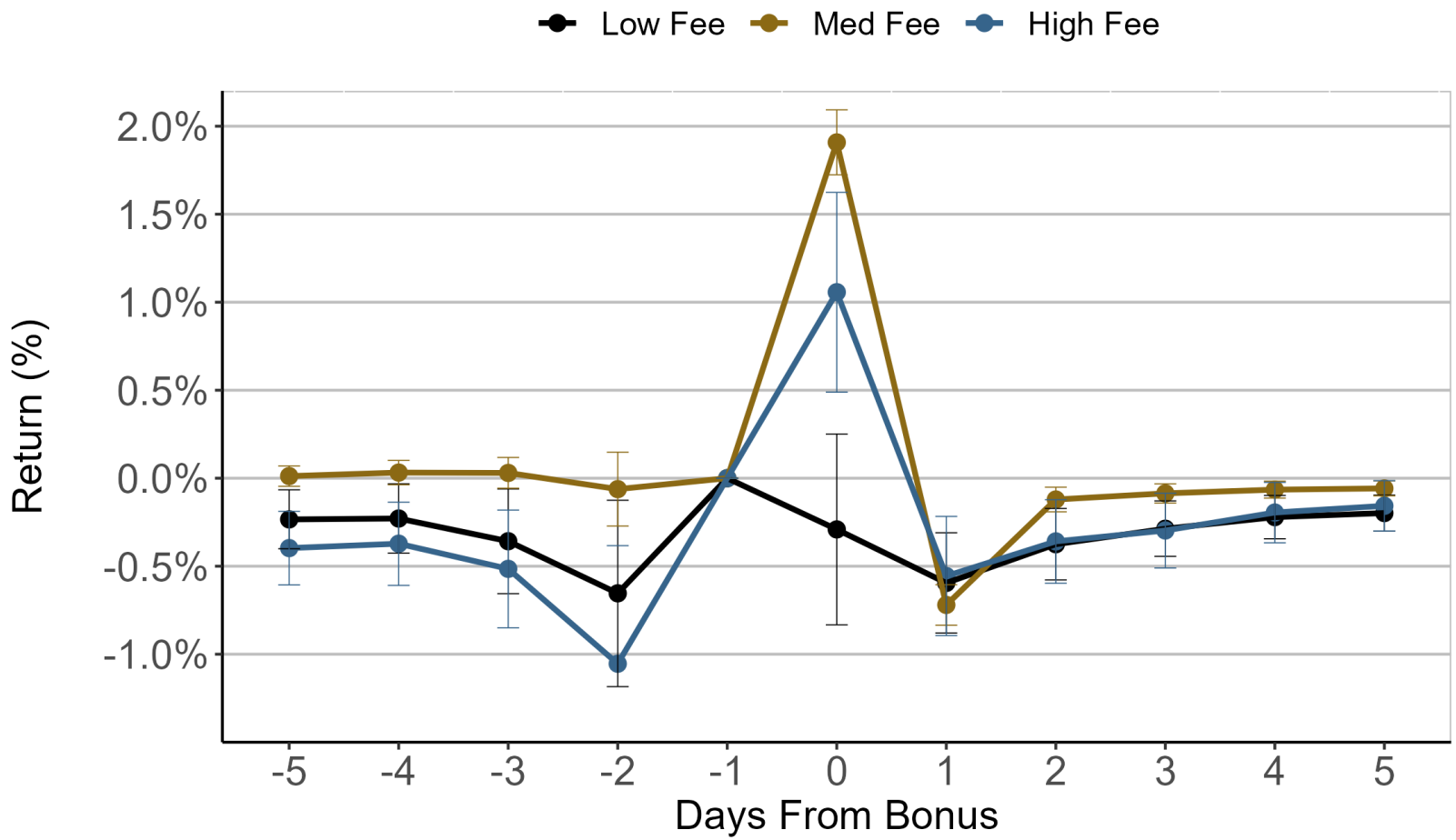


Noisy, yet positive, estimate of trades

Who Pushes Hardest for the Bonus?

- If the bonus drives the return jump, traders with the most to gain should push hardest
- A higher bonus rate means a larger payout from crossing the HWM
- Test: re-run the DiD around bonuses separately for low, medium, and high fee traders
- Prediction: the bonus-day return jump shows up in the higher-fee groups

Returns Around Bonuses By Performance Fee



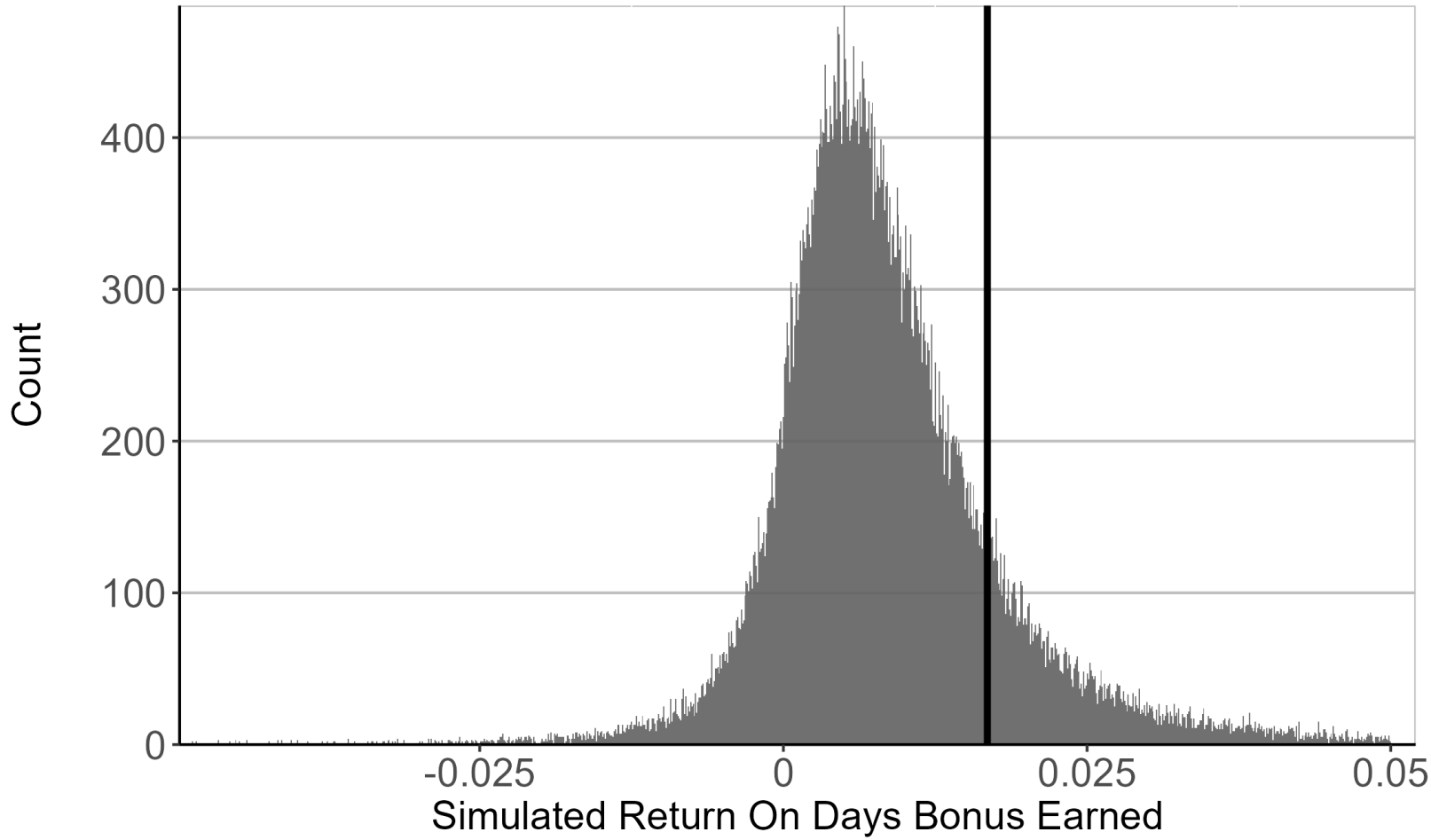
Higher fee traders have higher returns, consistent with incentives

▸ Number of Trades

Luck or Strategy?

- Returns must be positive on a bonus day, so how big is a 1.6 pp jump?
- Luck vs. skill bootstrap (Kosowski et al. 2006; Fama and French 2010)
- For each bonus day:
  - Keep the fund's portfolio weights from the prior day
  - Draw that day's asset returns at random from the assets held by all funds crossing that day
  - Repeat 10,000 times
- Compare the simulated returns to the actual bonus-day return

Bootstrap Return Distribution

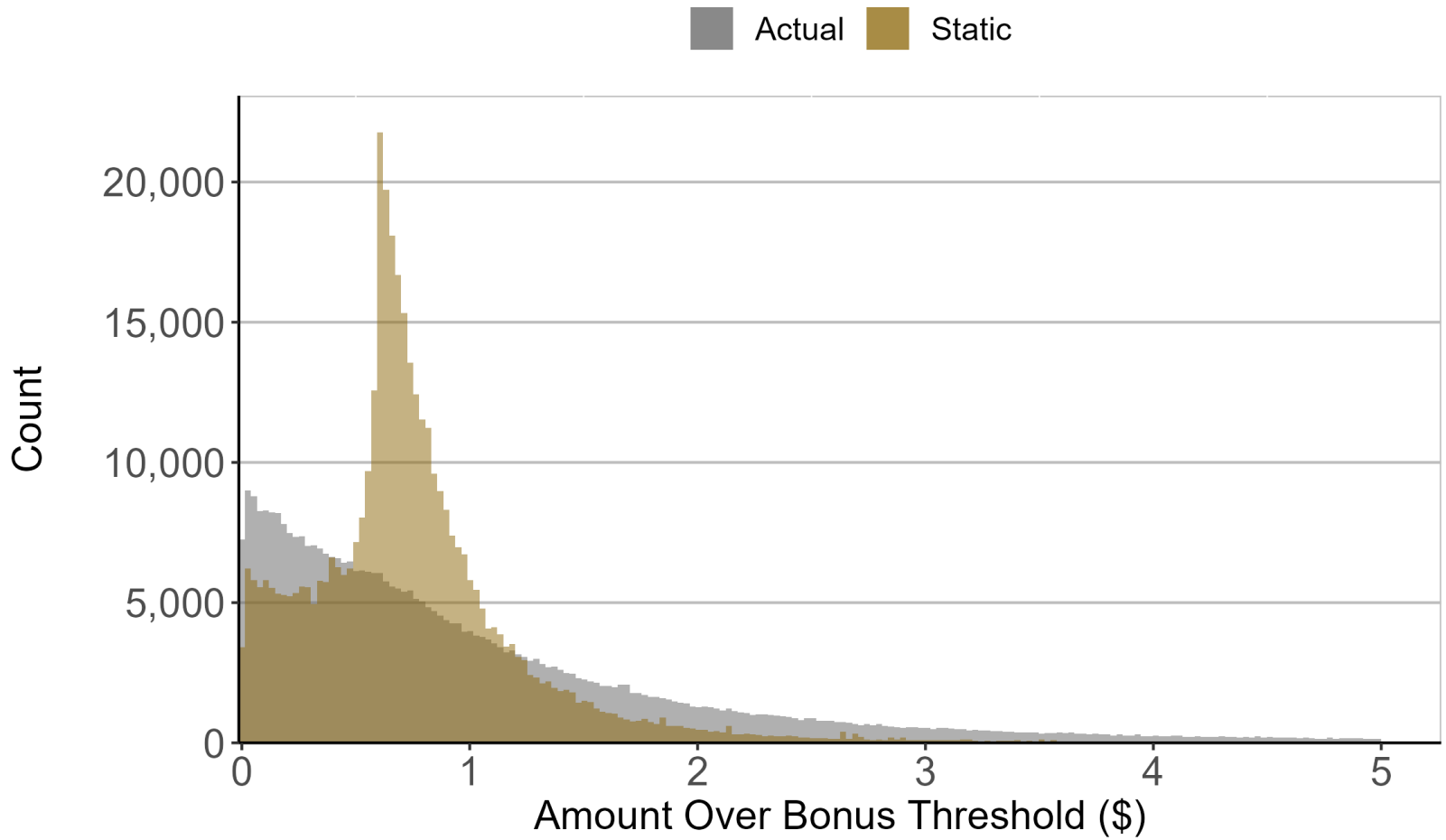


Observed 1.6% return is larger than 82% of simulations

Static Portfolio Counterfactual

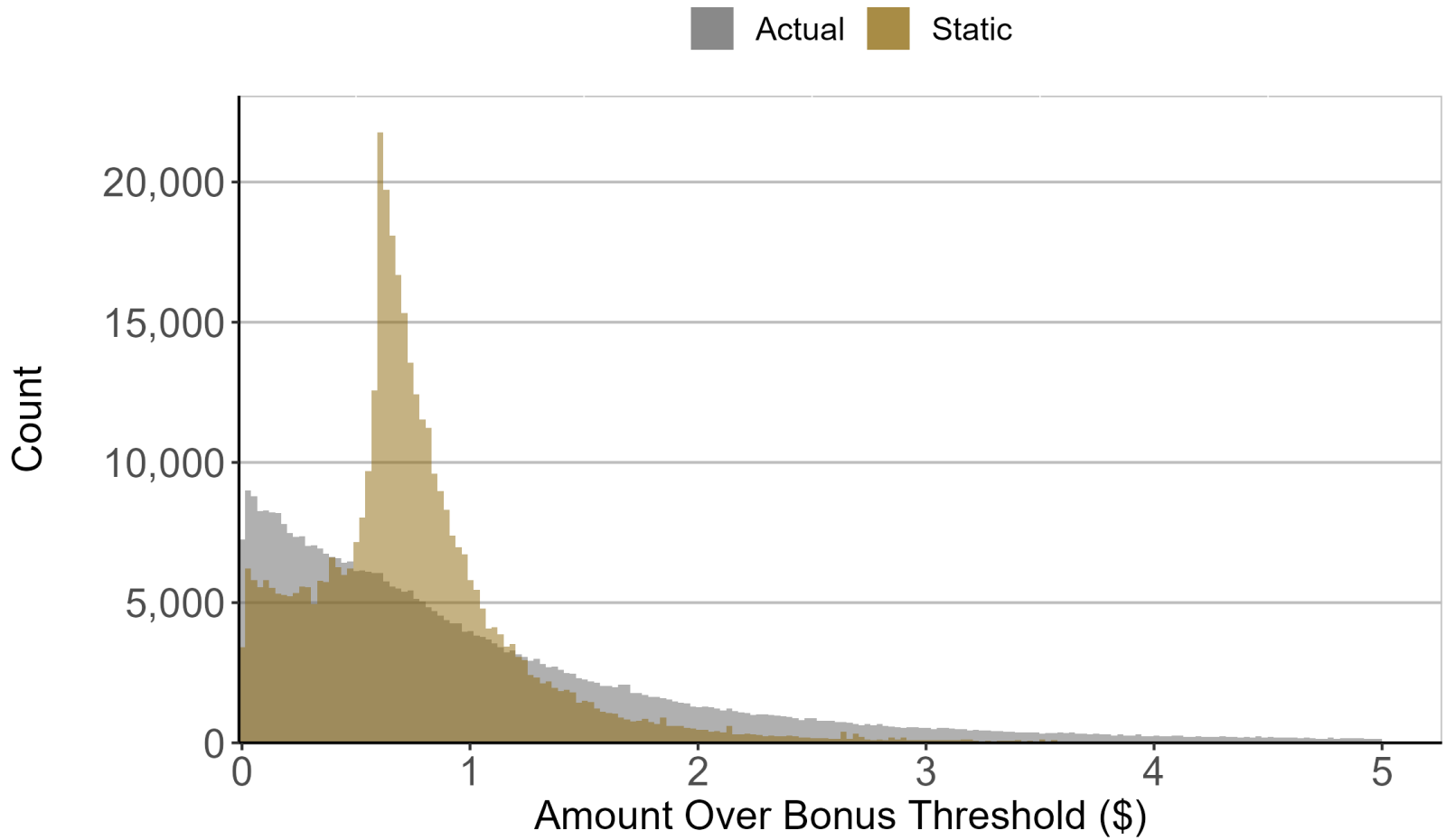
- What if traders did not trade at all?
- Freeze each fund's holdings on February 1st and hold them for a year
- Cumulative returns are similar and FF3 alpha is higher under a static setting for 63% of funds
- But is this optimal for bonus achievement?

Distance Over Bonus Threshold



Under the static trading strategy 57% of fund-years achieve more bonuses

Distance Over Bonus Threshold



Long right tail of actual crossings, consistent with HWM incentive

Back of the Envelope

- Is chasing the bonus worth it for the trader?
- **Gain:** the bonus as a share of fund value, $Bonus_{i,t}/Price_{i,t}$
- **Cost:** abnormal return, the next 10-day return after the bonus minus the 10-day return just before it

$$Net_{i,t} = \frac{Bonus_{i,t}}{Price_{i,t}} + \left(R_{i,t}^{+10} - R_{i,t}^{-10} \right)$$

- $Net_{i,t} < 0$: the push for the bonus cost more than it paid
- Average net loss of 2.5% of fund value; median of -1.4%
  - Statistically different from zero; robust to window length and to overlapping bonus windows

Summary of Additional Results

- **Mechanism** ▸ Details
  - Assets traded on bonus days had higher past 10-day returns (sales 7.4% vs. 4.8% for purchases)
  - Sold assets went on to earn more: 0.69 pp higher future returns than purchases
  - Sales jump from about 40% to 48% of trades on bonus days
- **Investors (new data, preliminary)** ▸ Details
  - Certificate trades rise about 25% on bonus days (top 20% most-traded)
  - Volume rises too, but the estimate is noisy

Conclusion

- Spike in returns corresponds with selling high-return assets and forgoing potential further gains
- High-frequency incentives can be counterproductive for retail investors
  - Exacerbating short-term biases and undermining sustainable performance
- Highlights importance of incentive frequency and interaction with agent type

Thanks!

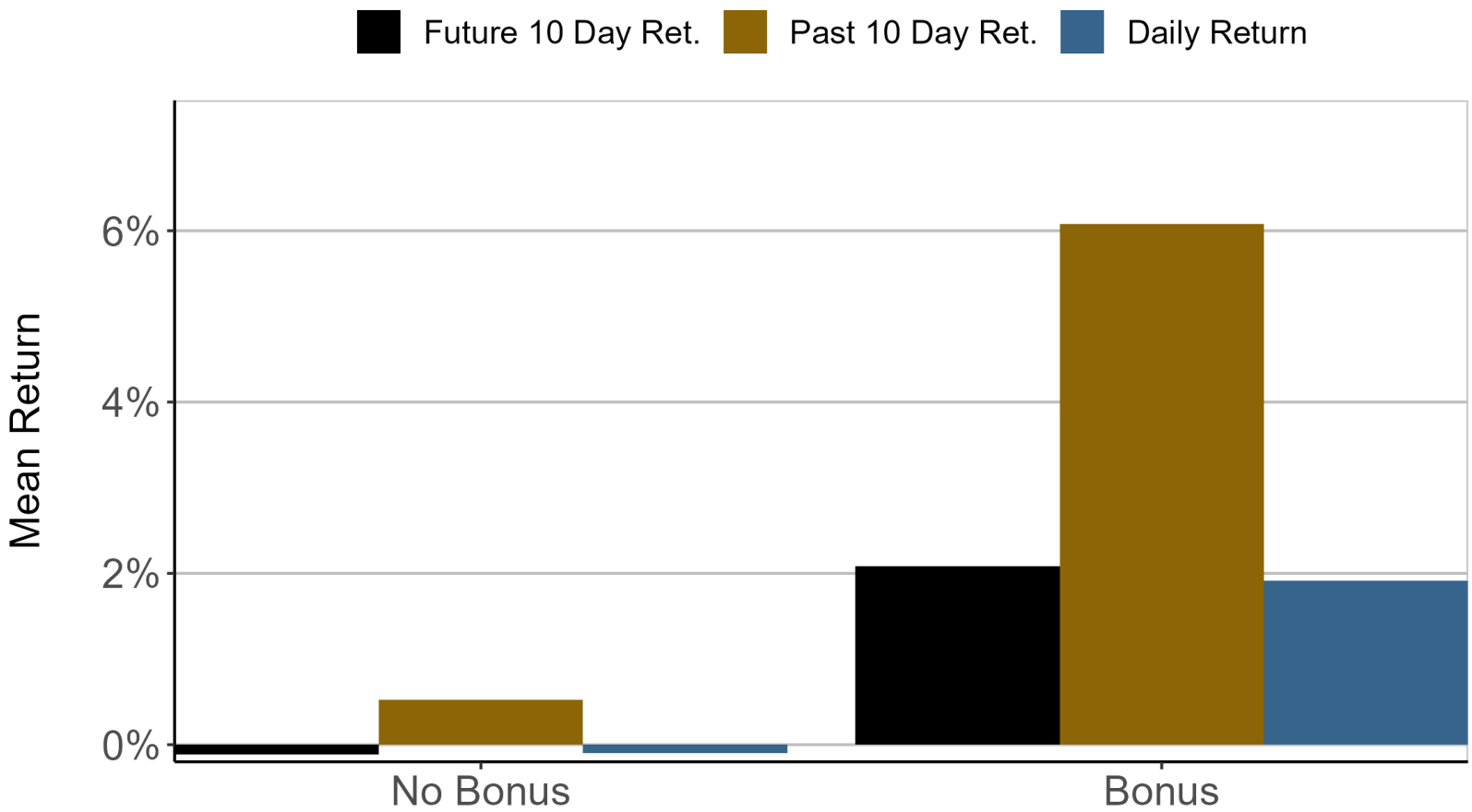
jjmohr1@olemiss.edu

www.jmohr.me

Mechanism: What Do Traders Trade?

- Use transaction-level data
- Compare assets bought and sold on bonus vs. non-bonus days
- Look at each asset's past 10-day, same-day, and next 10-day return

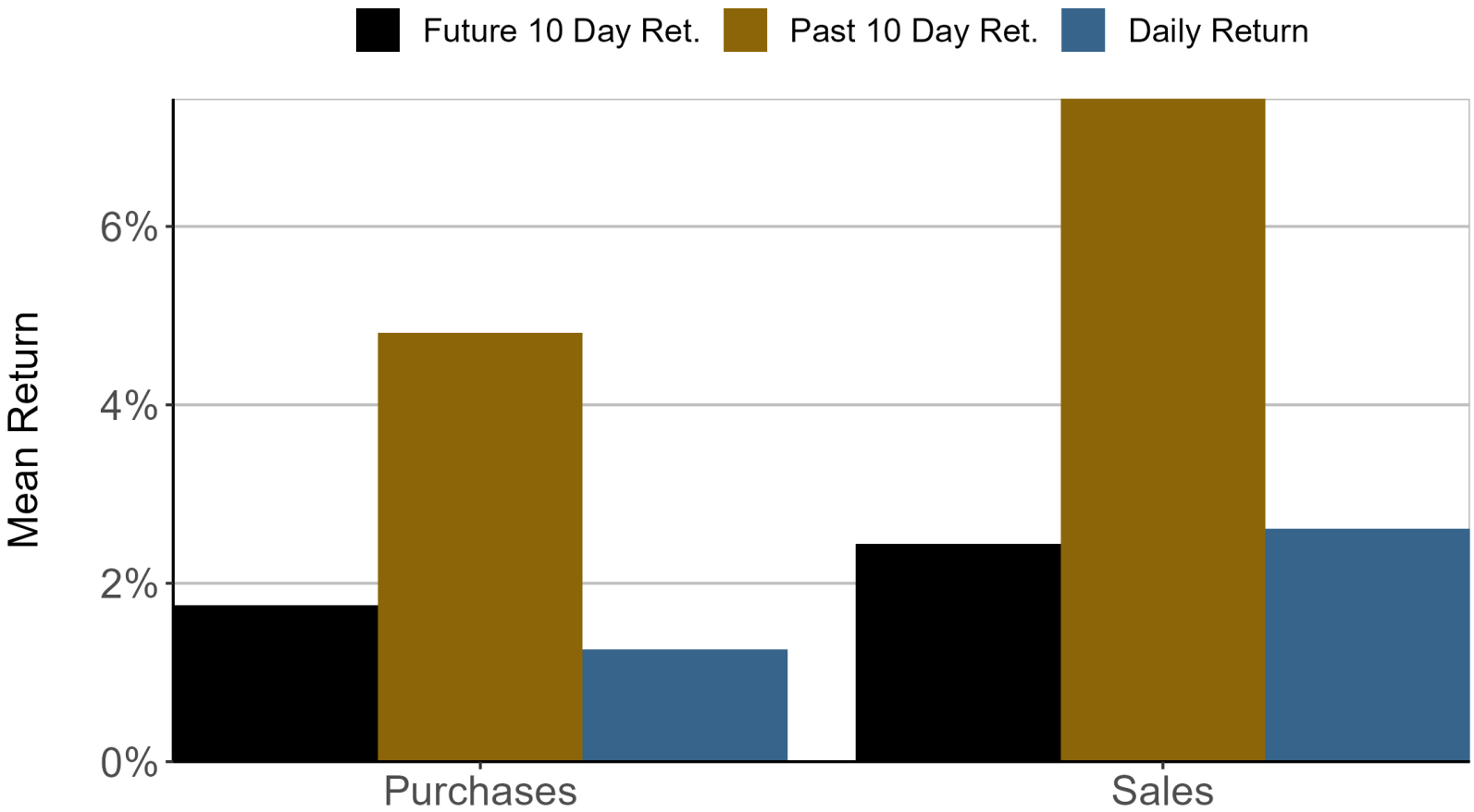
Return Patterns: Bonus vs. Non-Bonus Days



Assets traded on bonus days had higher past returns

▸ Sales Table ▸ Purchases Table

Return Patterns: Purchases vs. Sales on Bonus Days



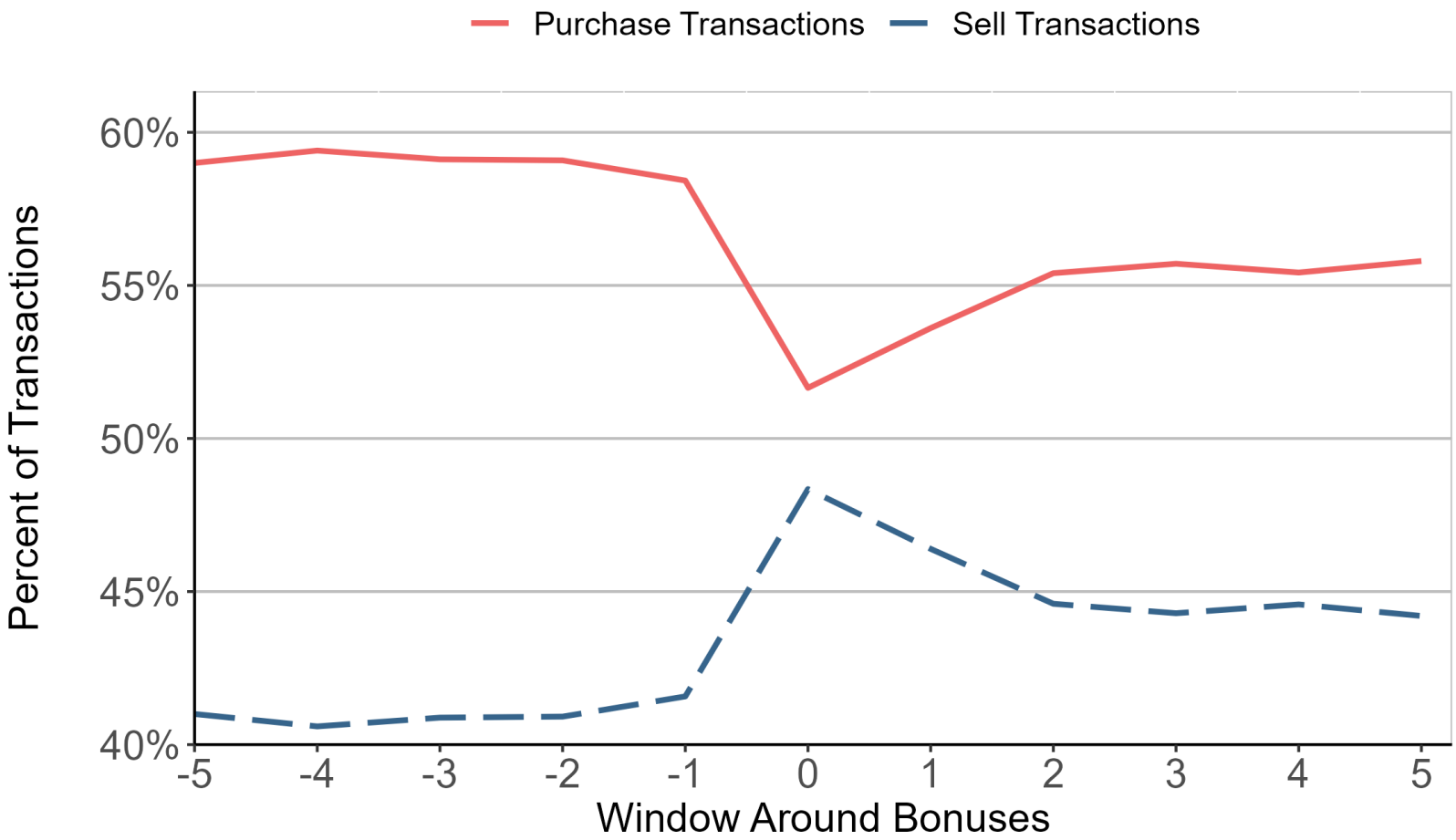
Future returns are larger for assets that traders sold on bonus days

▸ Sales Table ▸ Purchases Table

Selling to Lock In the Bonus

- Are traders selling recent winners to secure the bonus?
- Share of transactions that are sales, in the 10-day window around a bonus

Buying vs. Selling Activity



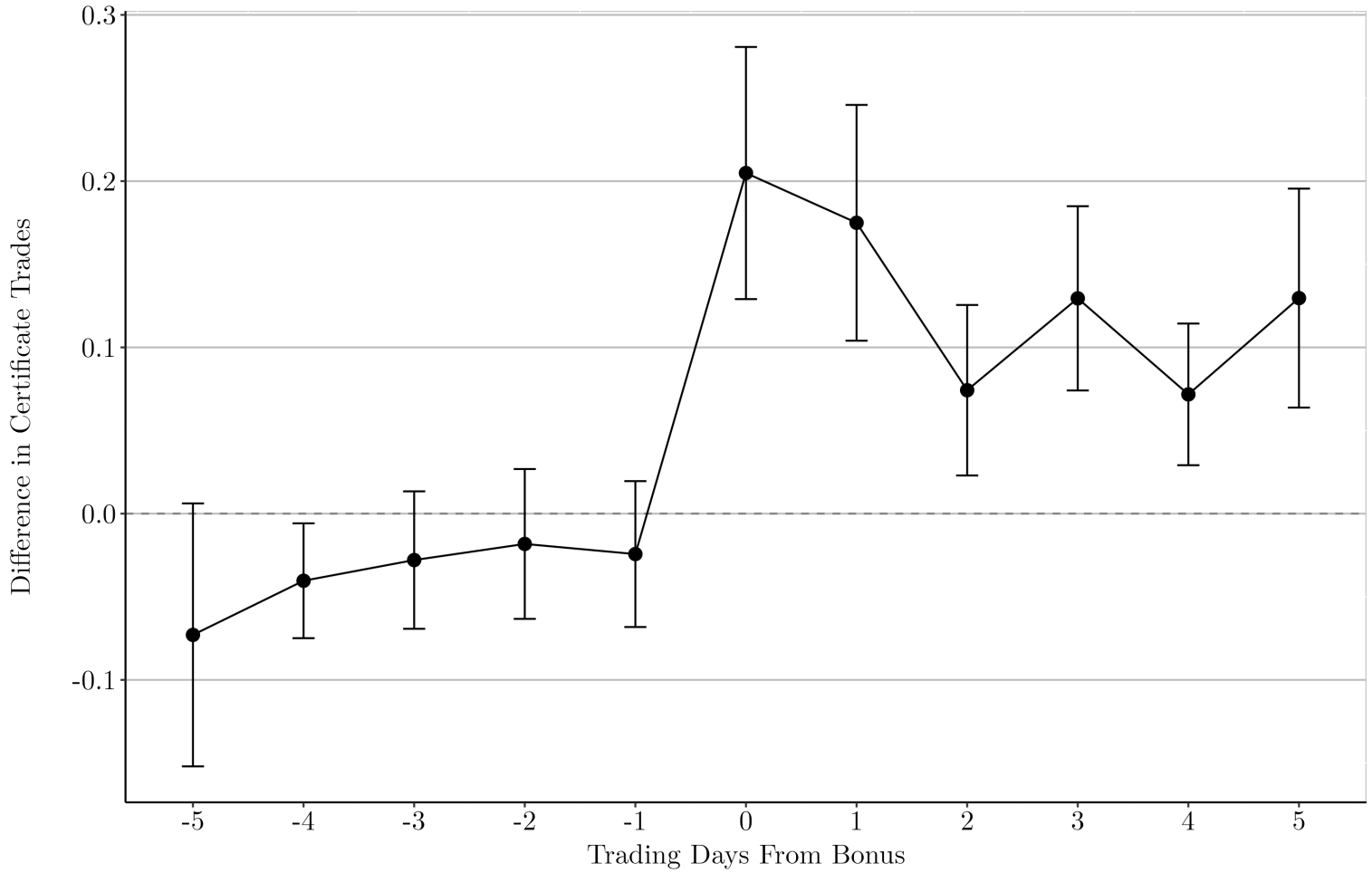
Consistent with selling recent winners to lock in the close price

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New Data: How Investors Trade the Certificates

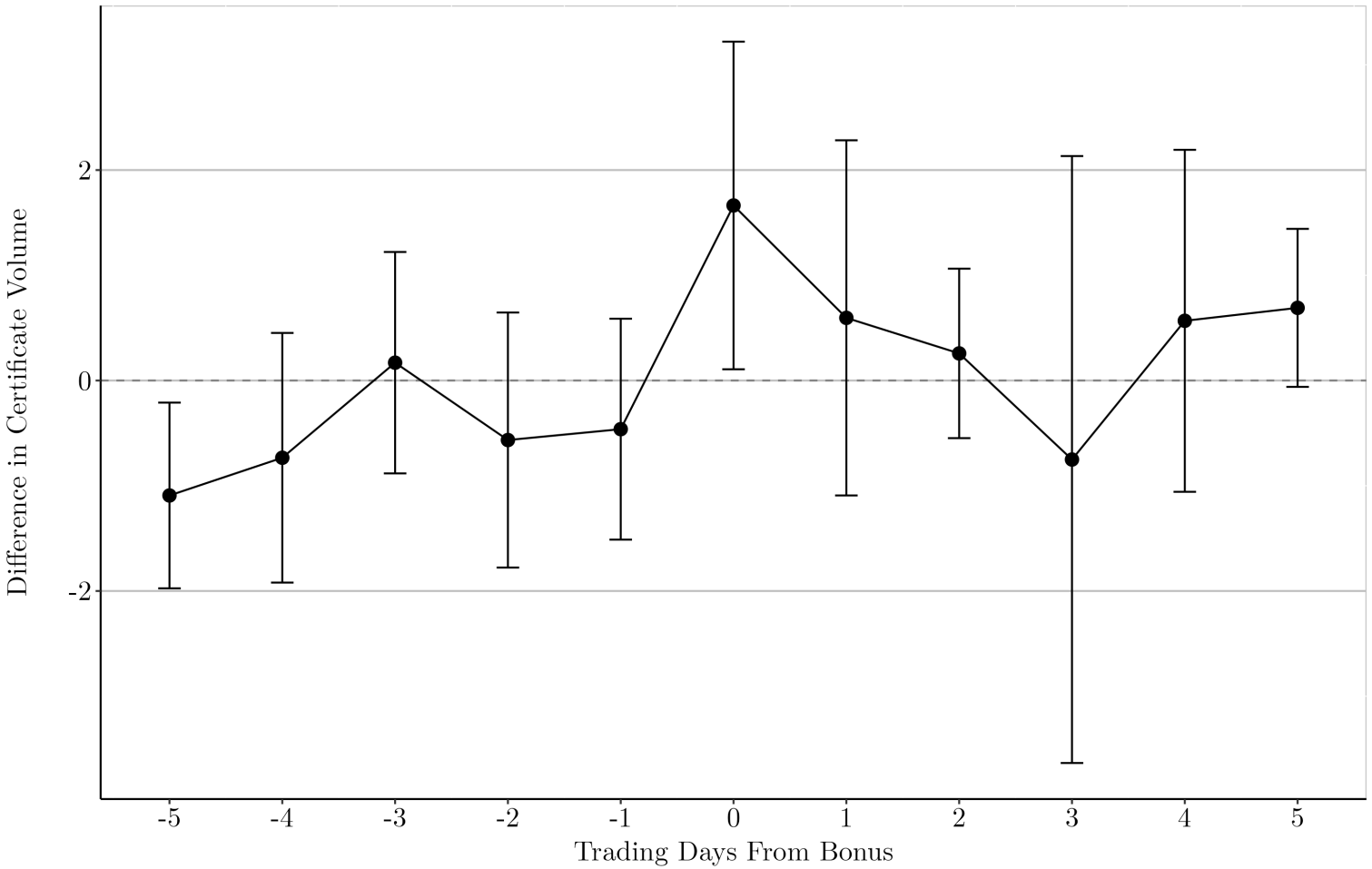
- Newly collected certificate trading data (preliminary results)
- Do investors respond to bonus days?
- Event study of certificate trades and volume around bonuses, in the most-traded certificates

Certificate Trades Around Bonuses: Top 20% Most-Traded



∼25% of the 0.82 daily mean for these certificates

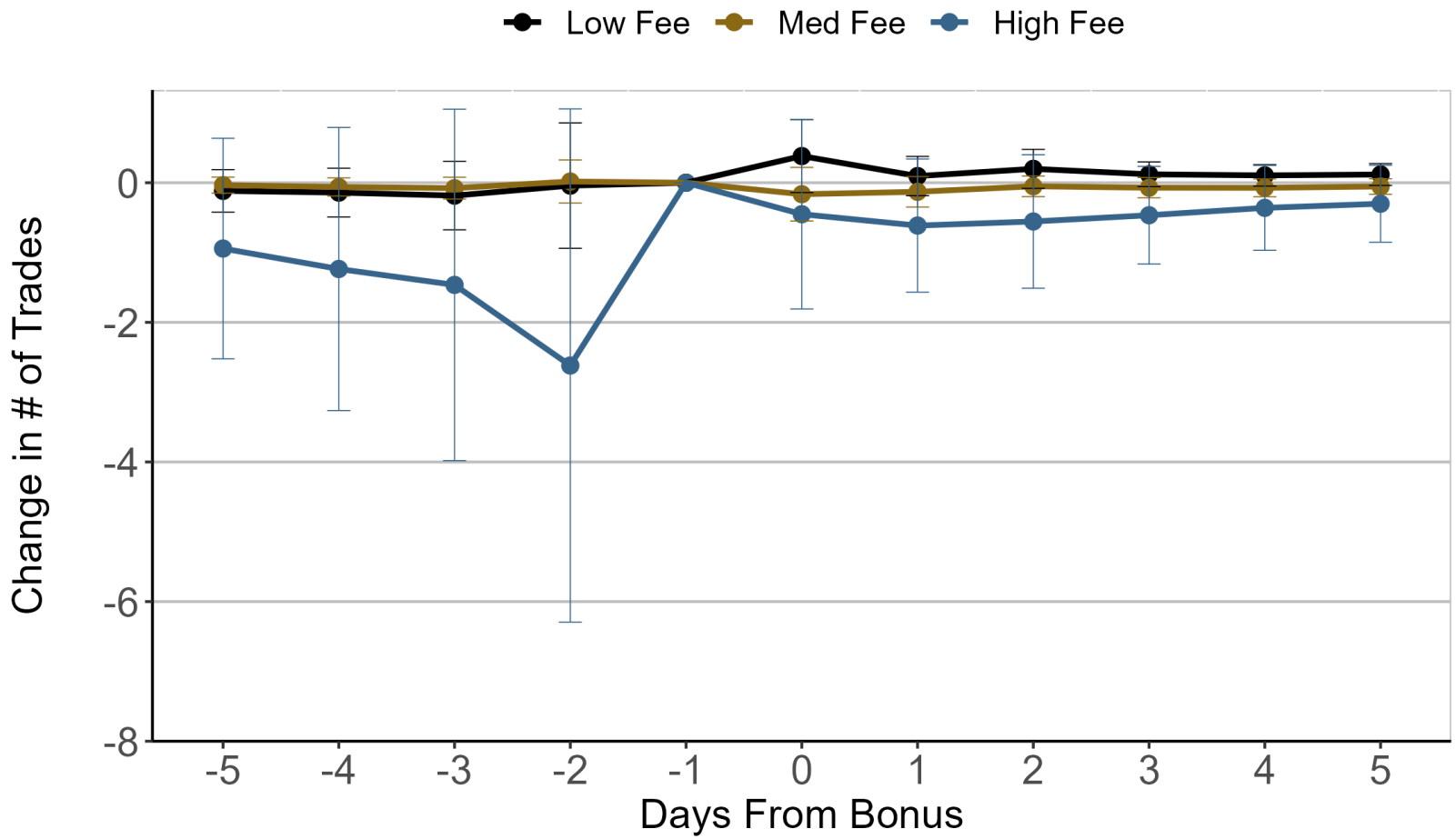
Certificate Volume Around Bonuses



∼23% of the 7.29 daily mean for these certificates

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Number of Trades Around Bonuses By Performance Fee



No difference across fee

◂ Returns By Performance Fee

RDD Around Bonus Achievement

	Number of Trades	Probability of Trading
	(1)	(2)
Intercept	3.237***	-0.497***
	(0.037)	(0.007)
I(Distance to Bonus $\geq$ 0)	0.312***	0.140***
	(0.051)	(0.009)
Distance to Bonus	-41.158***	-17.497***
	(6.745)	(1.244)
I(Distance to Bonus $\geq$ 0)$\times$ Distance to Bonus	58.004***	17.217***
	(9.845)	(1.811)
Observations	626,187	626,187
Adjusted R^2	0.0001	

◂ Number of Trades

Bonus Achievement Difference in Differences

	Returns	Number of Trades	Volatility
	(1)	(2)	(3)
Day -5	-0.001**	-0.095**	0.003
	(0.0002)	(0.045)	(0.002)
Day -4	-0.001***	-0.113**	0.004
	(0.0002)	(0.052)	(0.002)
Day -3	-0.001**	-0.163***	0.006
	(0.0003)	(0.061)	(0.003)
Day -2	-0.002**	-0.143	-0.003
	(0.001)	(0.103)	(0.004)
Day 0	0.016***	0.051	0.004
	(0.001)	(0.127)	(0.004)
Day +1	-0.006***	-0.091	0.001
	(0.0004)	(0.073)	(0.003)
Day +2	-0.002***	-0.063	0.0002
	(0.0003)	(0.053)	(0.002)
Day +3	-0.001***	-0.064	0.001
	(0.0002)	(0.051)	(0.003)
Day +4	-0.001***	-0.074	-0.001
	(0.0002)	(0.044)	(0.002)
Day +5	-0.001***	-0.059	-0.001
	(0.0002)	(0.039)	(0.002)
Observations	3,250,182	3,250,182	3,250,182

◂ Returns ◂ Number of Trades

Transactions When Bonus Achieved: Sales

Dependent Variables:	Next 10 Day Returns		Previous 10 Day Returns	
	(1)	(2)	(5)	(6)
Bonus Achieved	0.012***	0.007***	0.045***	0.021***
	(0.001)	(0.001)	(0.001)	(0.001)
Sale Transaction	-0.002*	-0.001	0.004**	0.005***
	(0.001)	(0.001)	(0.001)	(0.001)
Bonus Achieved × Sale Transaction	0.005***	0.004***	0.022***	0.020***
	(0.001)	(0.001)	(0.001)	(0.001)
Previous 10 Day Returns	0.129***	0.121***		
	(0.003)	(0.003)		
Daily Return			0.022***	0.020***
			(0.005)	(0.004)
Fixed-effects				
Fund	Yes	Yes	Yes	Yes
Date		Yes		Yes
Observations	3,620,016	3,620,016	3,635,974	3,635,974
Adjusted R^2	0.026	0.021	0.025	0.011

◂ Bonus Vs Non Bonus Days ◂ Purchases Vs Sales

Transactions When Bonus Achieved: Purchases

Dependent Variables:	Next 10 Day Returns		Previous 10 Day Returns	
	(3)	(4)	(7)	(8)
Bonus Achieved	0.017***	0.011***	0.067***	0.041***
	(0.001)	(0.001)	(0.002)	(0.001)
Buy Transaction	0.002*	0.000	-0.004**	-0.005***
	(0.001)	(0.001)	(0.001)	(0.001)
Bonus Achieved × Buy Transaction	-0.005***	-0.004***	-0.022***	-0.020***
	(0.001)	(0.001)	(0.001)	(0.001)
Previous 10 Day Returns	0.129***	0.121***		
	(0.003)	(0.003)		
Daily Return			0.022***	0.020***
			(0.005)	(0.004)
Fixed-effects				
Fund	Yes	Yes	Yes	Yes
Date		Yes		Yes
Observations	3,620,016	3,620,016	3,635,974	3,635,974
Adjusted R^2	0.026	0.021	0.025	0.011

◂ Bonus Vs Non Bonus Days ◂ Purchases Vs Sales